

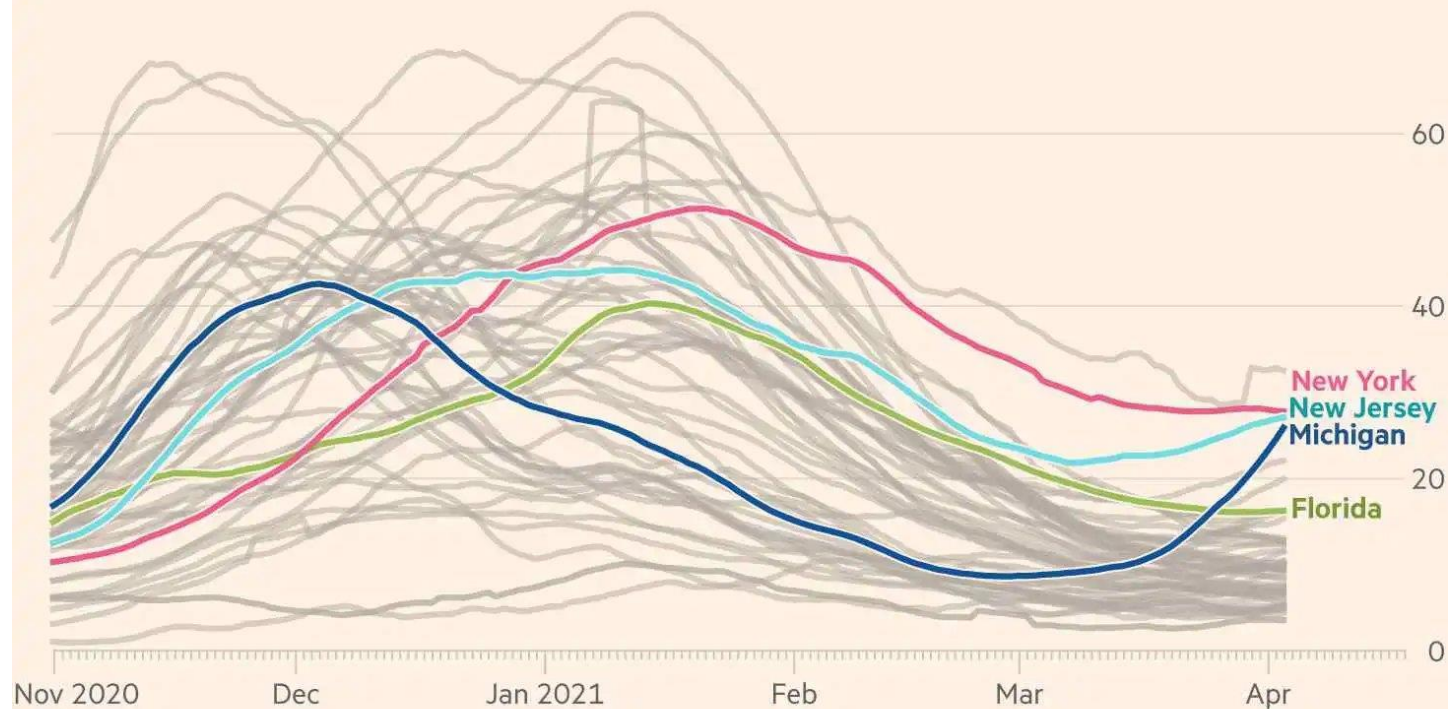
Visualizing Uncertainty

Alex Kale

Problems when reasoning with uncertainty

Michigan's Covid-19 hospitalisations are rising

Patients per 100,000 (7-day average)

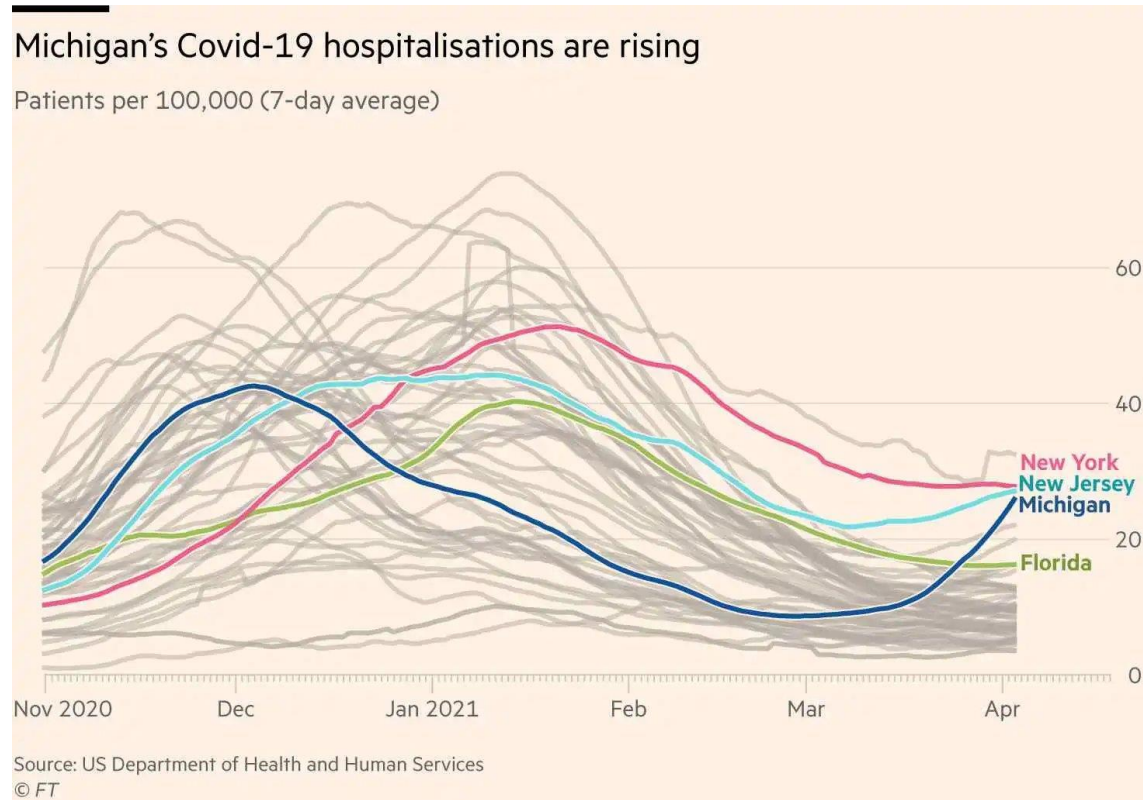


Source: US Department of Health and Human Services

© FT

Burn-Murdoch. *Financial Times*. 2021

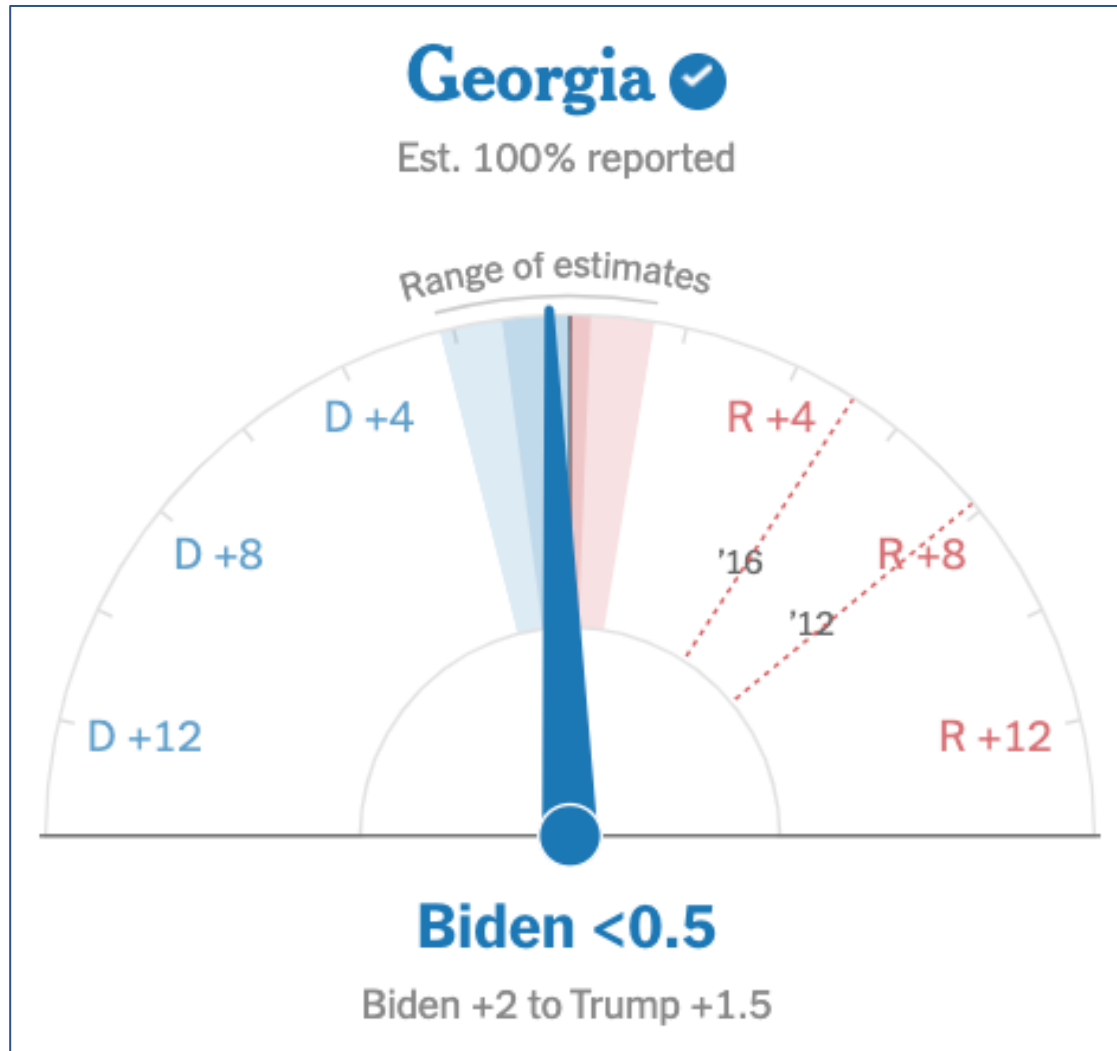
Problems when reasoning with uncertainty



Burn-Murdoch. *Financial Times*. 2021

Uncertainty representation implicit or missing

Problems when reasoning with uncertainty



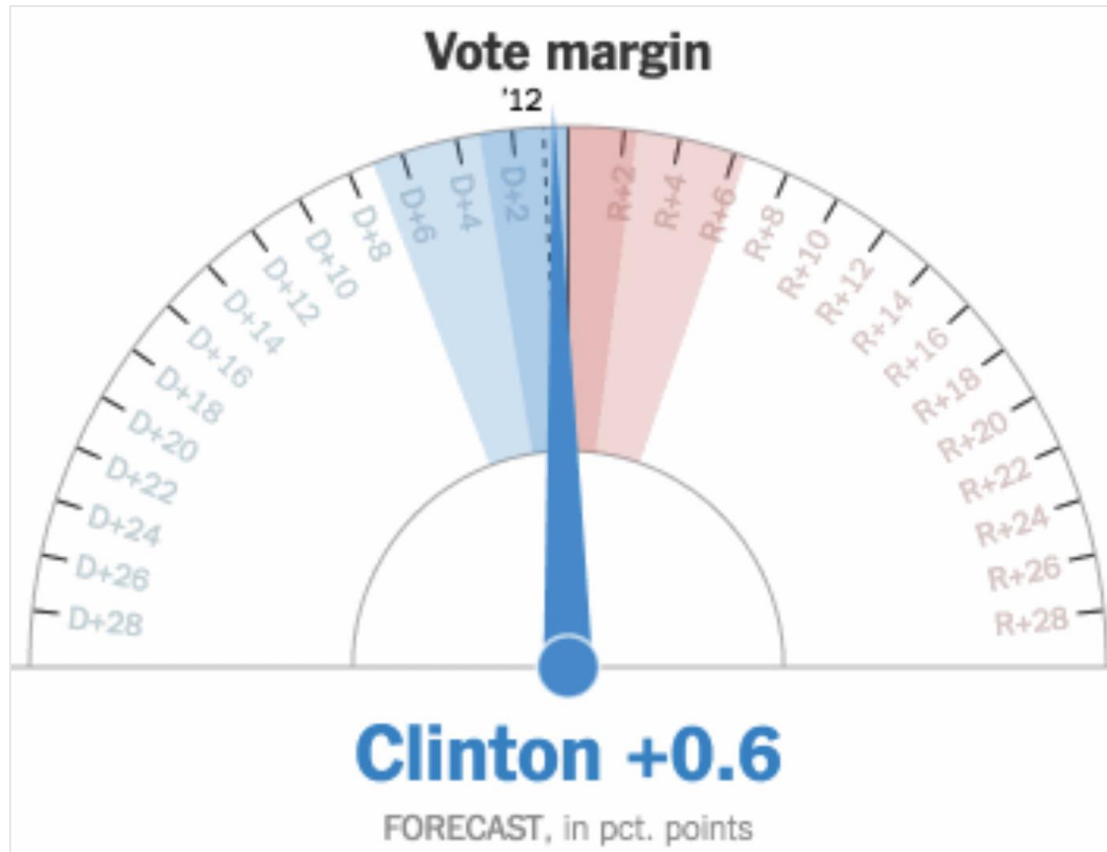
Hypothetical outcome plot (HOP)

Makes uncertainty visceral,
experienceable

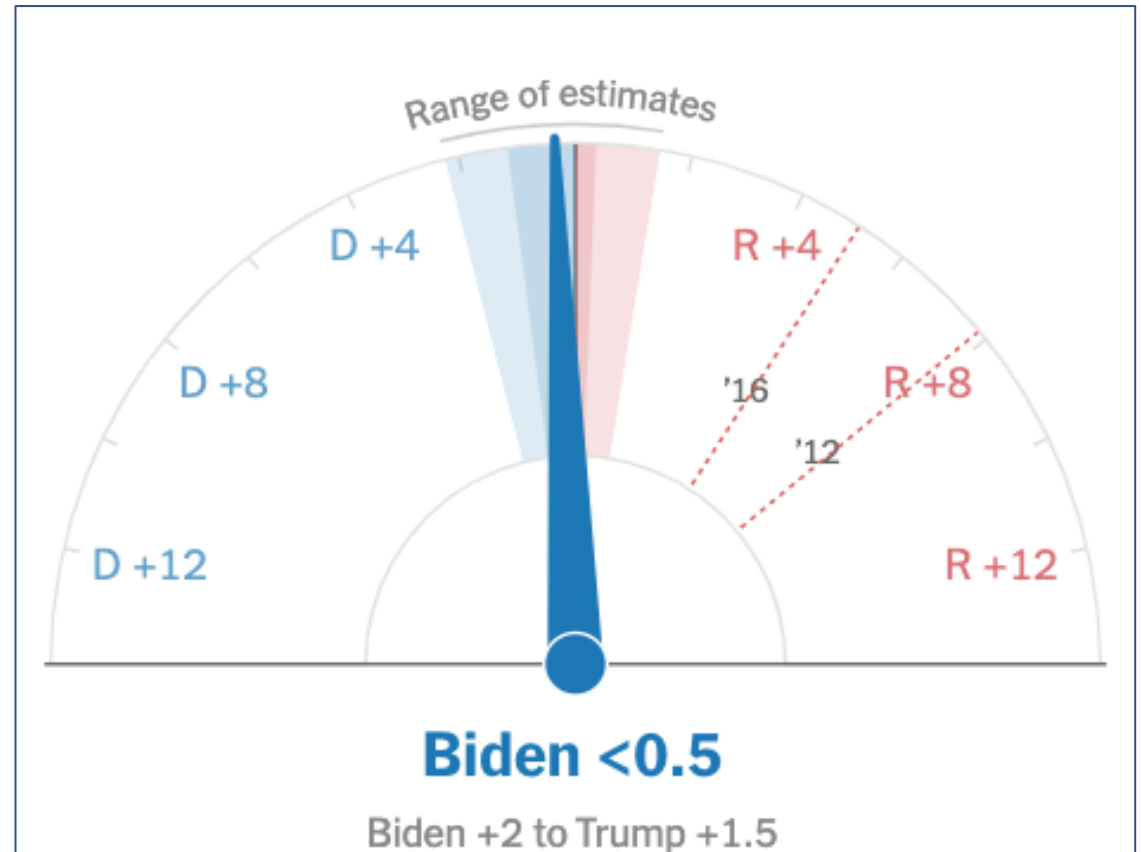
Kale, Nguyen, Kay, Hullman. HOPs Help Untrained Observers Judge Trends in Ambiguous Data. *IEEE VIS*. 2018.

Some called this “Irresponsible”

Problems when reasoning with uncertainty



New York Times. *Live Presidential Forecast*. 2016.

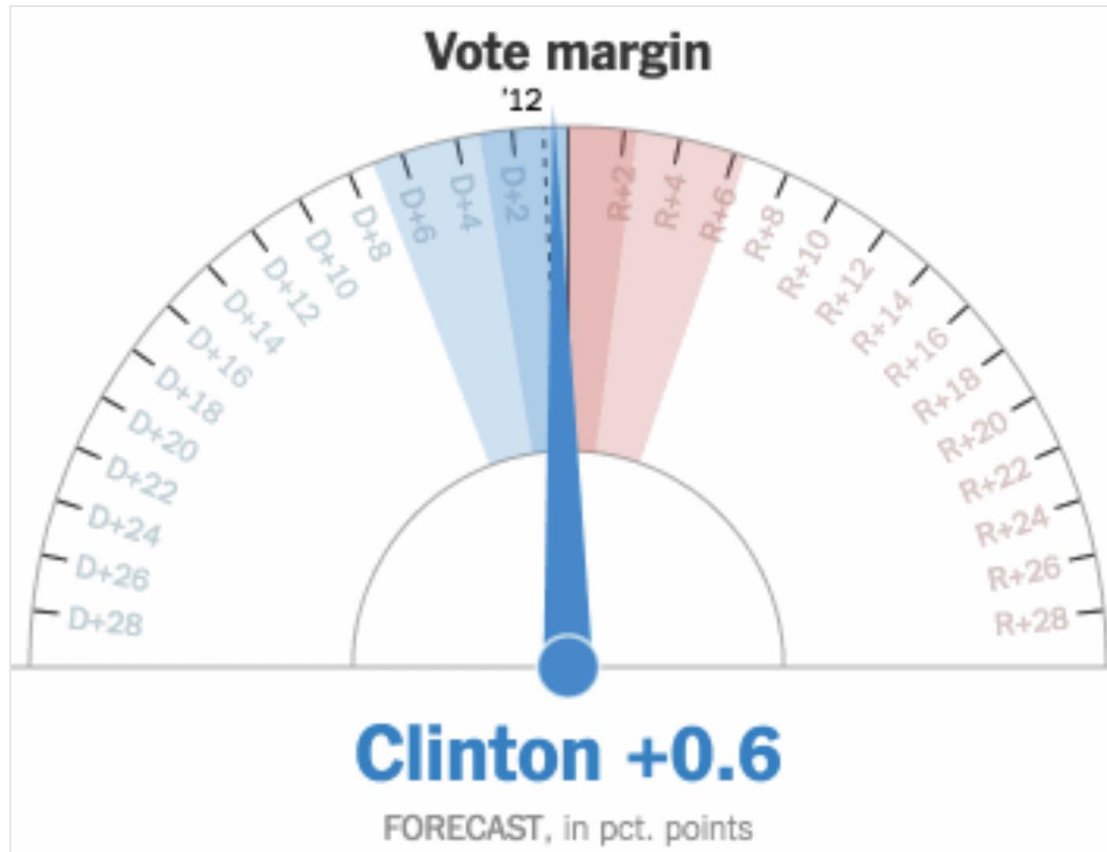


New York Times. *Live Presidential Forecast*. 2020.

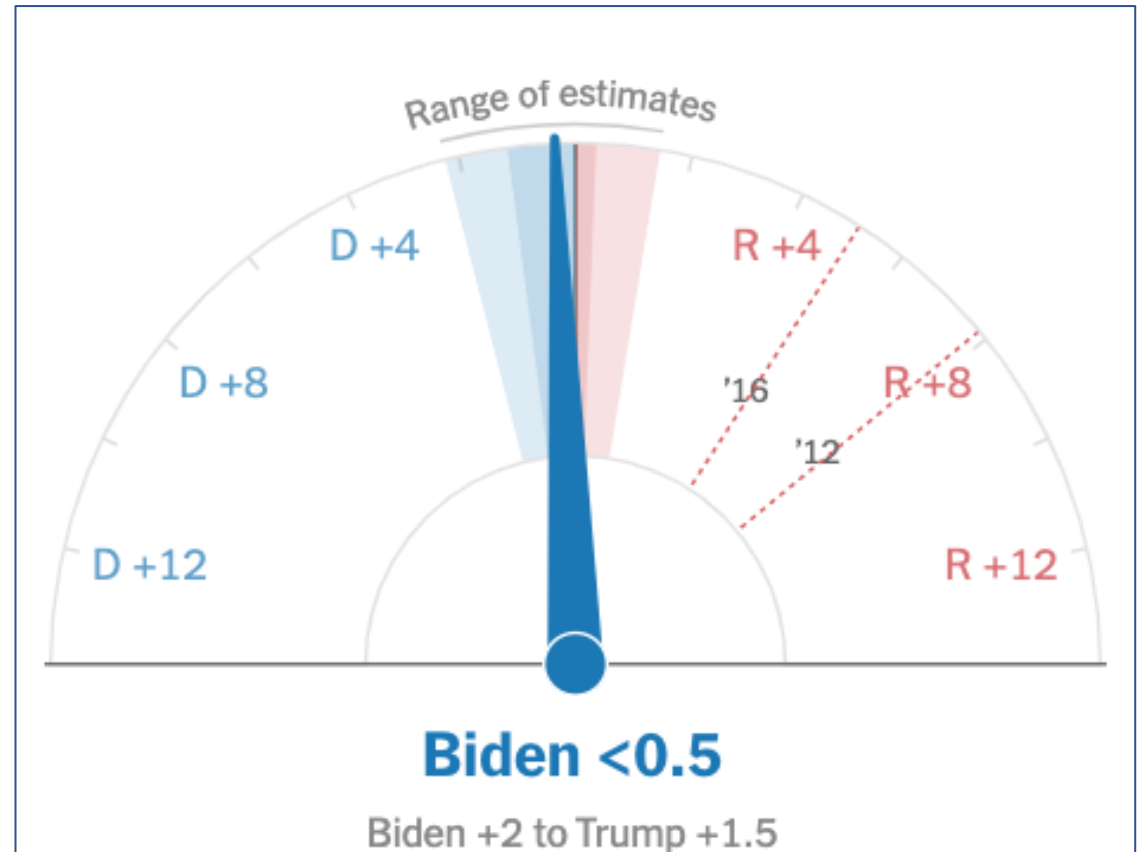
Uncertainty seen as an obstacle, confusing

Authors Don't Visualize Uncertainty? *IEEE VIS*. 2020.

Problems when reasoning with uncertainty



New York Times. *Live Presidential Forecast*. 2016.

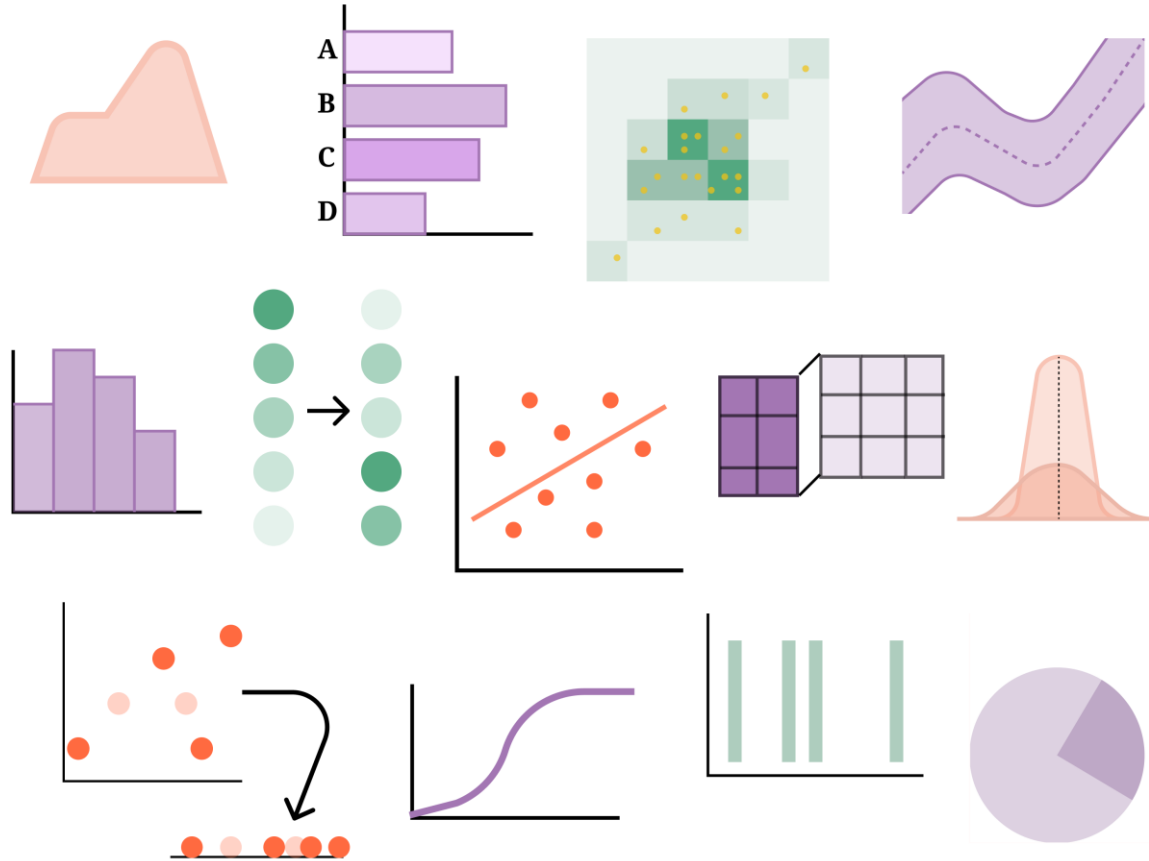


New York Times. *Live Presidential Forecast*. 2020.

Visual representation impacts interpretation

Gleicher. Error Bars Considered Harmful... *IEEE VIS*. 2014.

Problems when reasoning with uncertainty

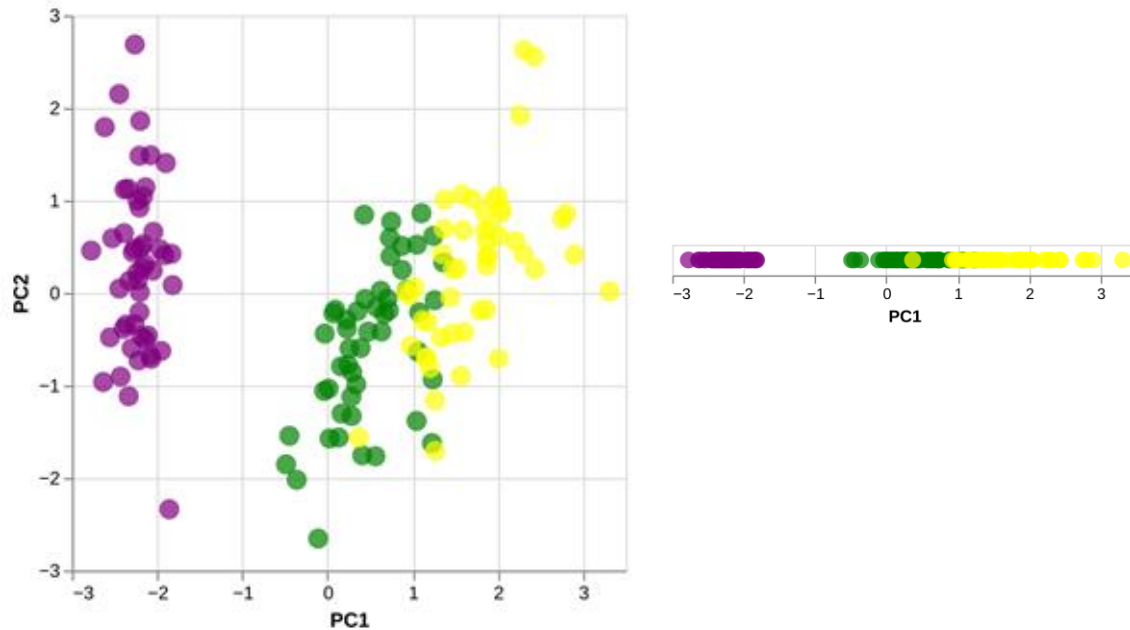


“All visualizations do dimensionality reduction”

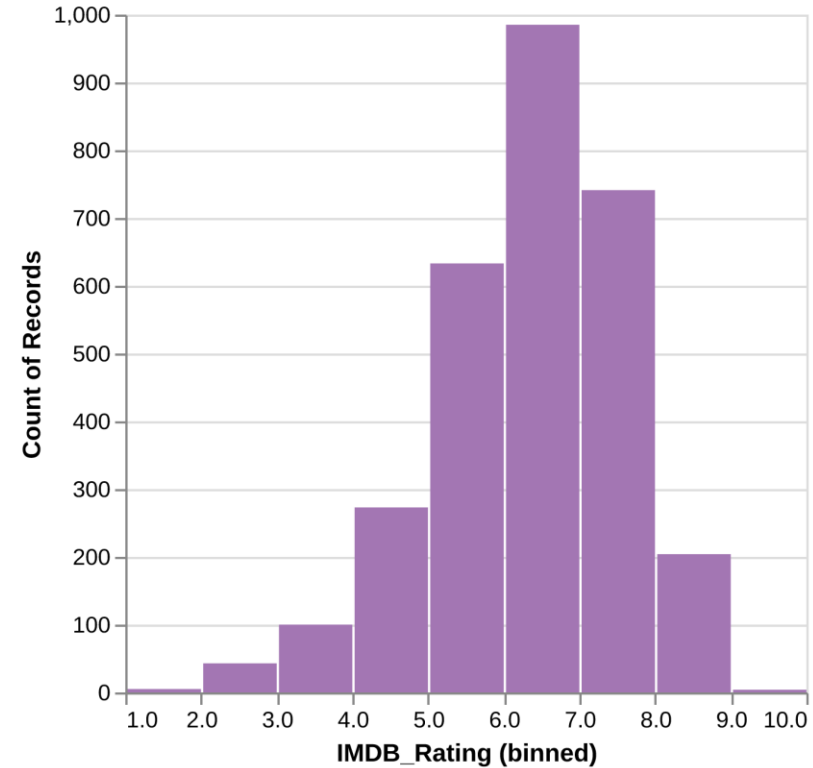
Visualizations hide a lot of information

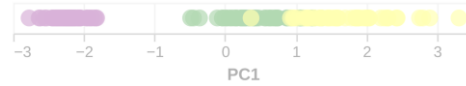
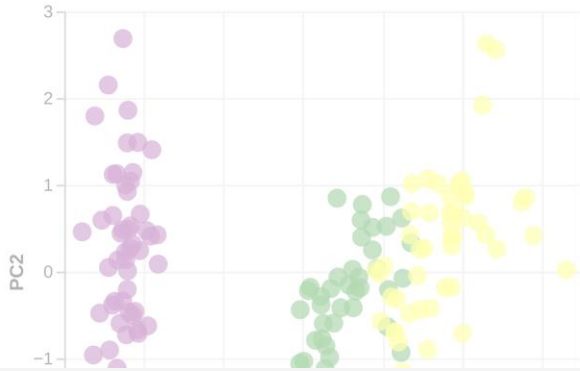
Mehta, Kindlmann, Kale. Designing for Disclosure in Data Visualization. *IEEE VIS*. 2026.

Some visualizations
represent dimensionality
reduction in its purest form



But for other visualizations,
it is not that obvious





```
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import pandas as pd
import altair as alt
```

```
# Loading the data
iris = load_iris()
X = iris['data']
y = iris['target']
```

```
# Transforming the data
X_std = StandardScaler().fit_transform(X)
```

```
# Dimensionality reduction
pca = PCA(n_components=2).fit(X_std)
X_pca = pca.transform(X_std)
```

```
pca_data = pd.DataFrame({
    'PC1': X_pca[:, 0],
    'PC2': X_pca[:, 1],
    'target': y
})
```

```
# Visualizing the data
chart_1d = alt.Chart(pca_data).mark_circle(
    x='PC1:Q',
    color=alt.Color('target:N', scale=alt.Scale(
        'yellow']), legend=None),
)
```

```
chart_2d = alt.Chart(pca_data).mark_circle(size=120).encode(
    x='PC1:Q',
    y='PC2:Q',
    color=alt.Color('target:N', scale=alt.Scale(range=['purple', 'green',
    'yellow']), legend=None),
)
```

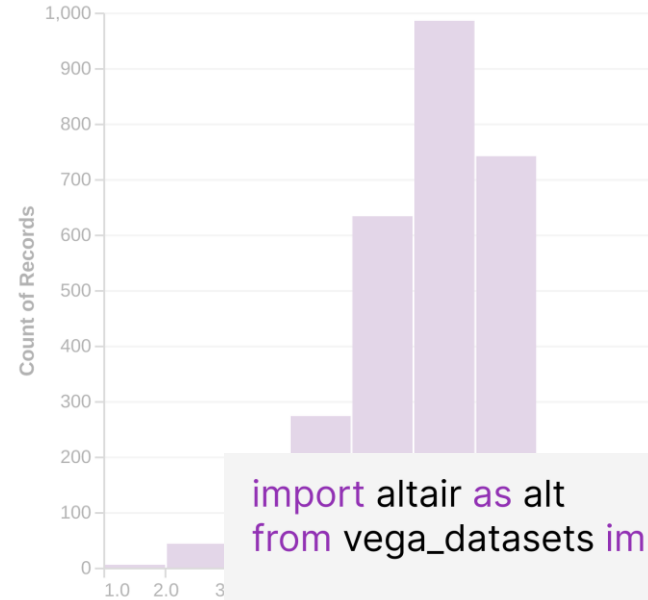
Transforming the data

```
X_std = StandardScaler().fit_transform(X)
```

Dimensionality reduction

```
pca = PCA(n_components=2).fit(X_std)
X_pca = pca.transform(X_std)
```

```
pca_data = pd.DataFrame({
    'PC1': X_pca[:, 0],
    'PC2': X_pca[:, 1],
    'target': y
})
```



```
import altair as alt
from vega_datasets import data
```

```
source = data.movies.url
```

```
chart = alt.Chart(source).mark_bar().encode(
    alt.X("IMDB_Rating:Q", bin=True),
    y='count()',
    color=alt.value('#A376B3')
)
```

Should we visualize uncertainty?

Visualizing uncertainty

“Uncertainty must be displayed if it is to be reasoned with and incorporated into the visual analytics process. In existing visualizations, much of the information is displayed as if it were ‘true.’”

(Thomas and Cook 2005)

Why should we show uncertainty?

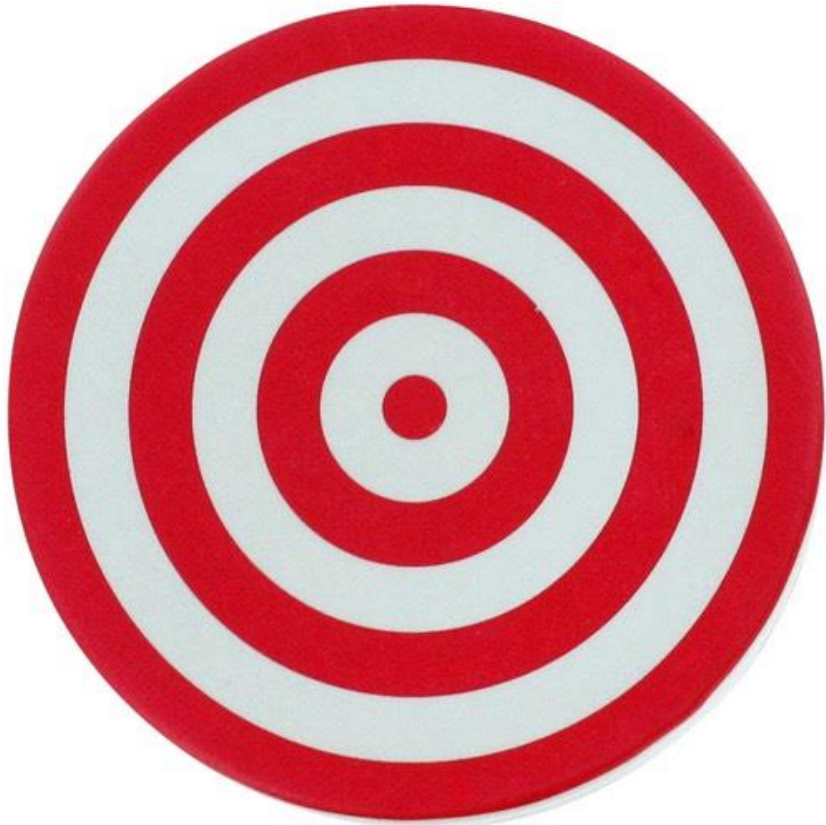
- Guard against overconfidence (i.e., ignoring uncertainty)
 - Insensitivity to sample size
 - Not acknowledging assumptions
- Visualizations often used for inferences & decisions
 - People may infer different amounts of uncertainty when omitted

Uncertainty:

“the possibility of multiple outcomes”

Terminology

Precision



Terminology

Precision



Terminology

Precision



Terminology

Precision



Accuracy



Terminology

Precision



Accuracy



Terminology

Precision

Variance



Accuracy

Bias



Terminology

Reliability

P1



P2



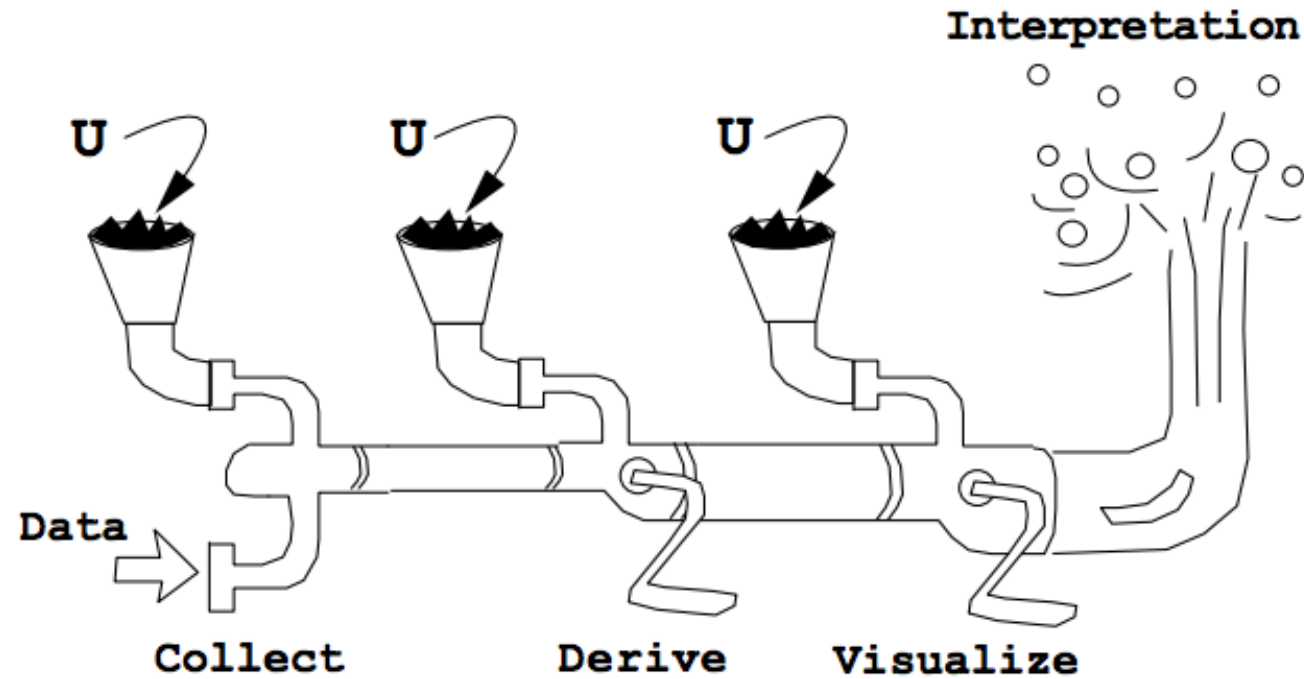


Figure 1: This visualization pipeline shows the introduction of uncertainty (U) from models and measurements, uncertainty from transformation processes, and uncertainty from the visualization process itself.

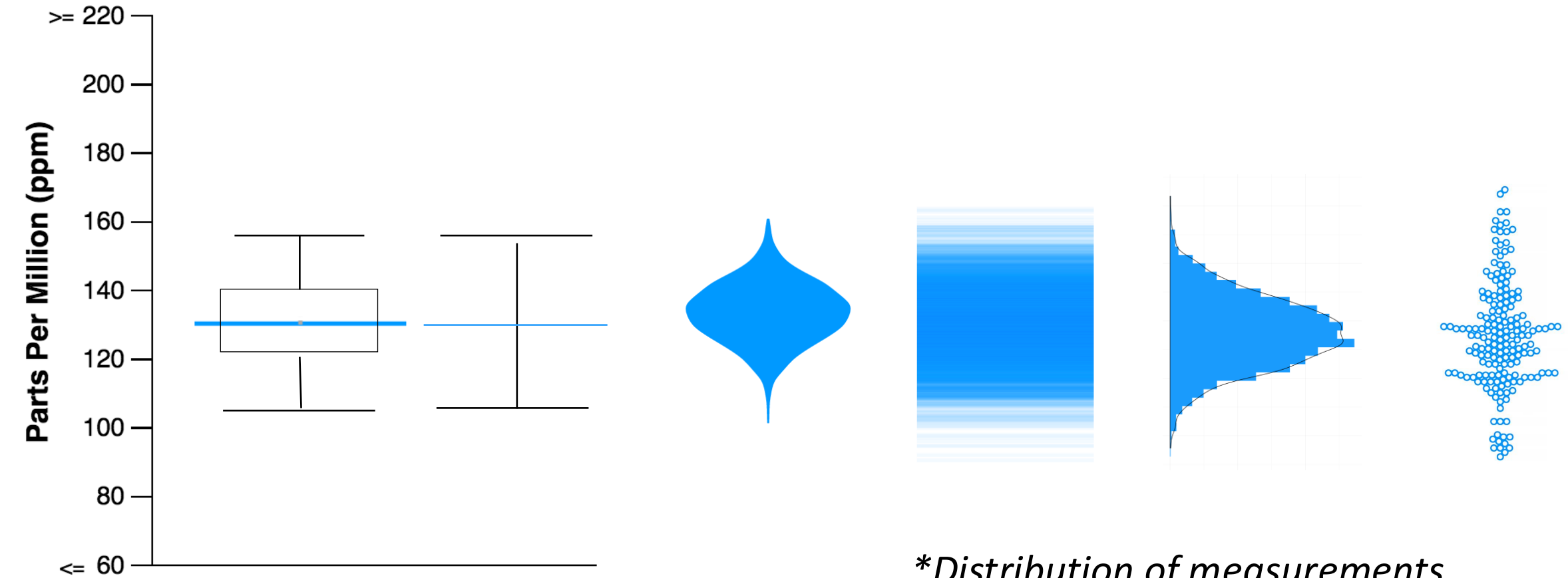
Example: Quantifying Uncertainty

Imagine that we took many measurements of the concentration of a solute in seawater.

Data (in *parts per billion*): [132.24, 131.72, 129.93, 128.60, 129.16, 127.76, 134.45, 127.03, 133.20, 133.64, 129.34, 131.96, 131.55, 131.76, 120.39, 127.49, ...]

Q: How much do these readings vary?

Visualize



Q: How much do these readings vary?

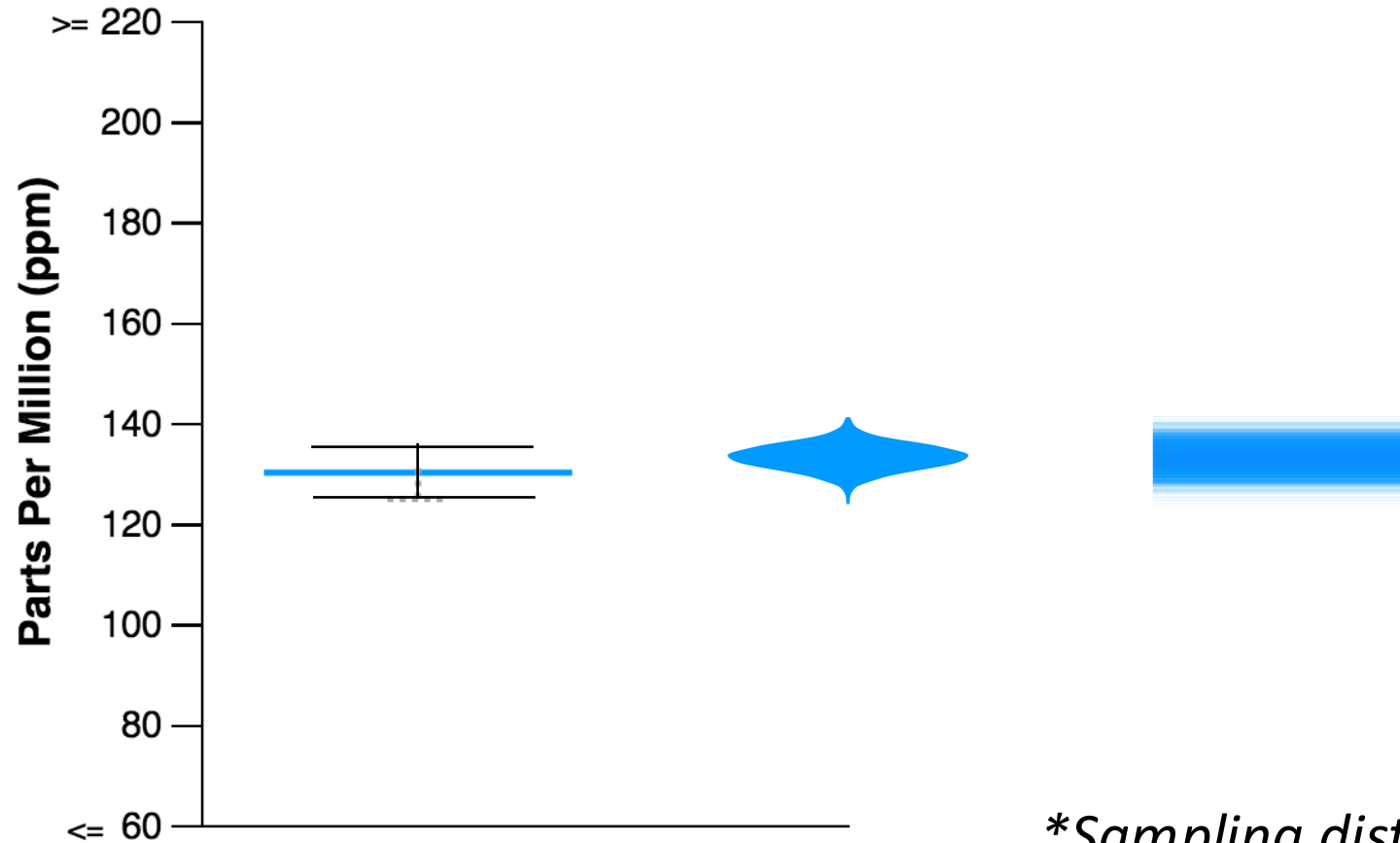
Example: Quantifying Uncertainty

Imagine that we took many measurements of the concentration of a solute in seawater.

Data (in *parts per billion*): [132.24, 131.72, 129.93, 128.60, 129.16, 127.76, 134.45, 127.03, 133.20, 133.64, 129.34, 131.96, 131.55, 131.76, 120.39, 127.49, ...]

Q: How does our estimation of the level in the entire ocean vary?

Visualize

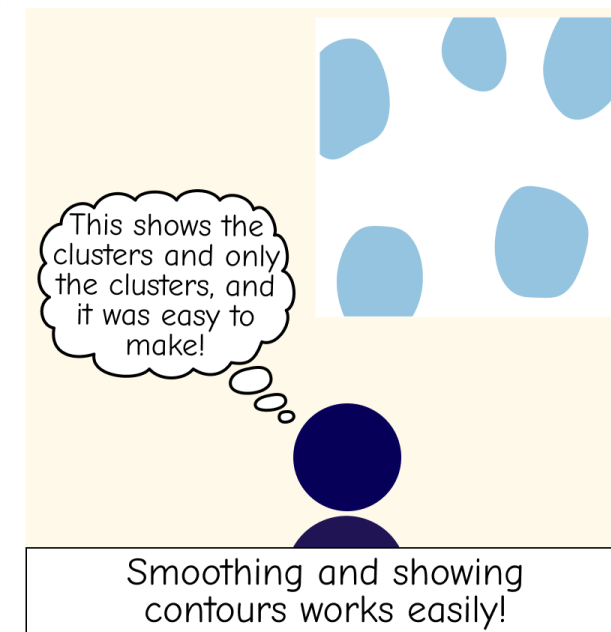
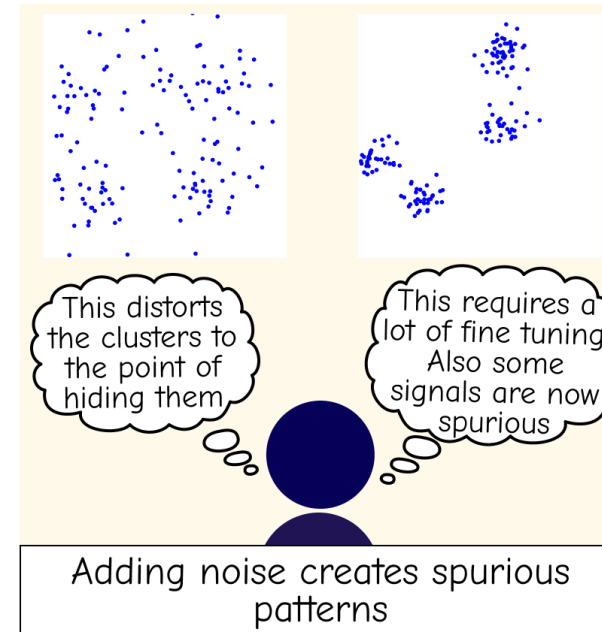
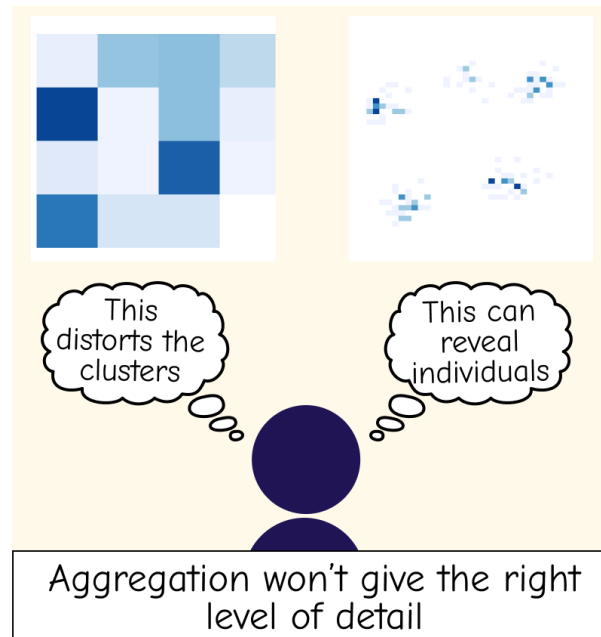
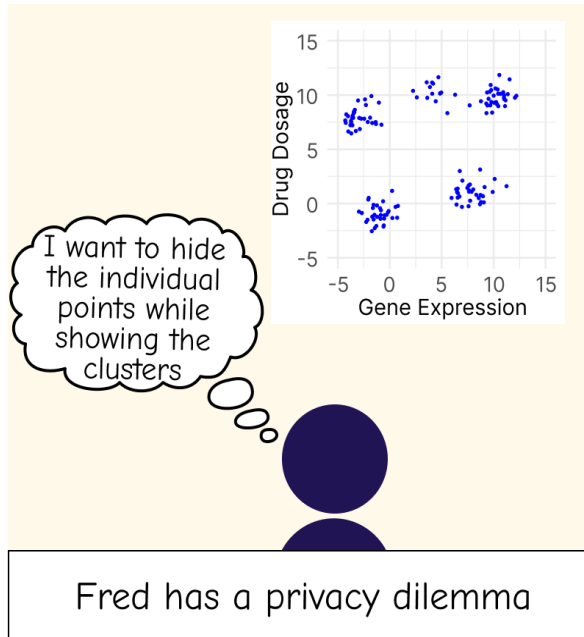


**Sampling distribution*

Q: How does our estimation of the level in the entire ocean vary?

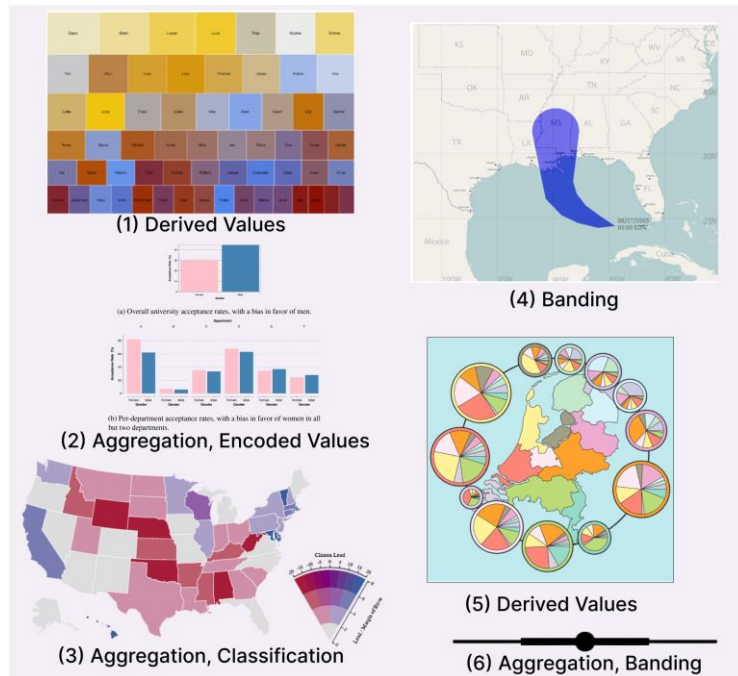
Example: Fred's Privacy Dilemma

Hide Individual Points
Show Clusters

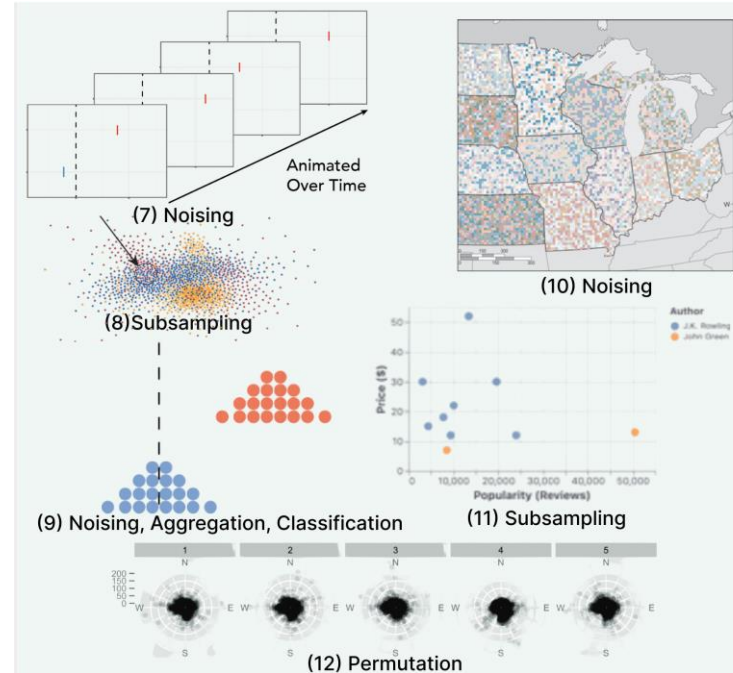


Three Families of Disclosure Tactics

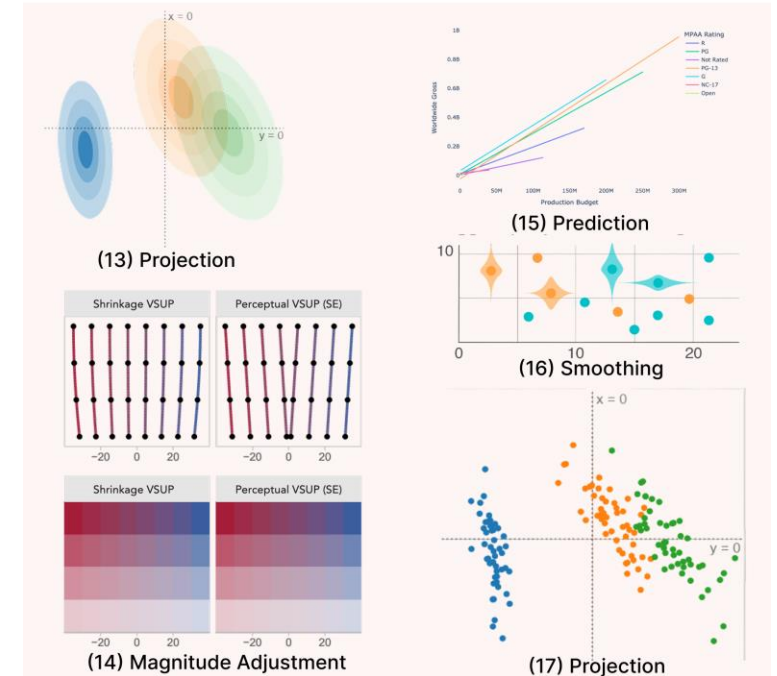
Summarizing



Sampling

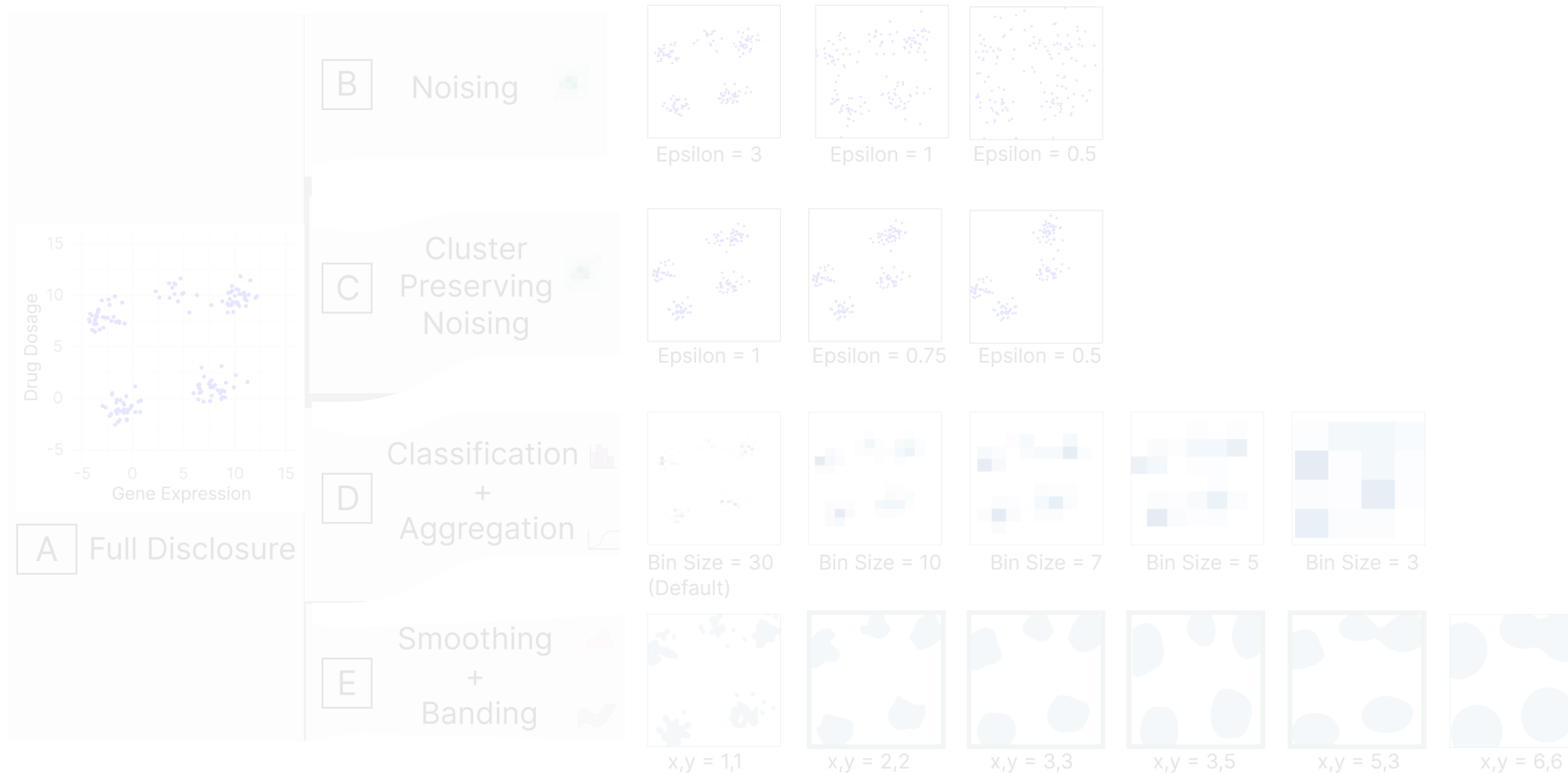


Modeling



Example: Fred's Privacy Dilemma

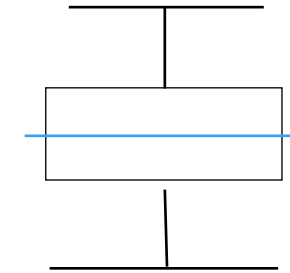
Hide Individual Points
Show Clusters



Visualization Approaches

Glyphs and summary plots

- Error bars
- Boxplots



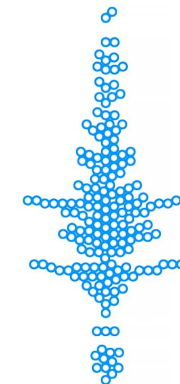
Visual variables

- Lightness/value, area
- Blur/focus, “sketchiness”



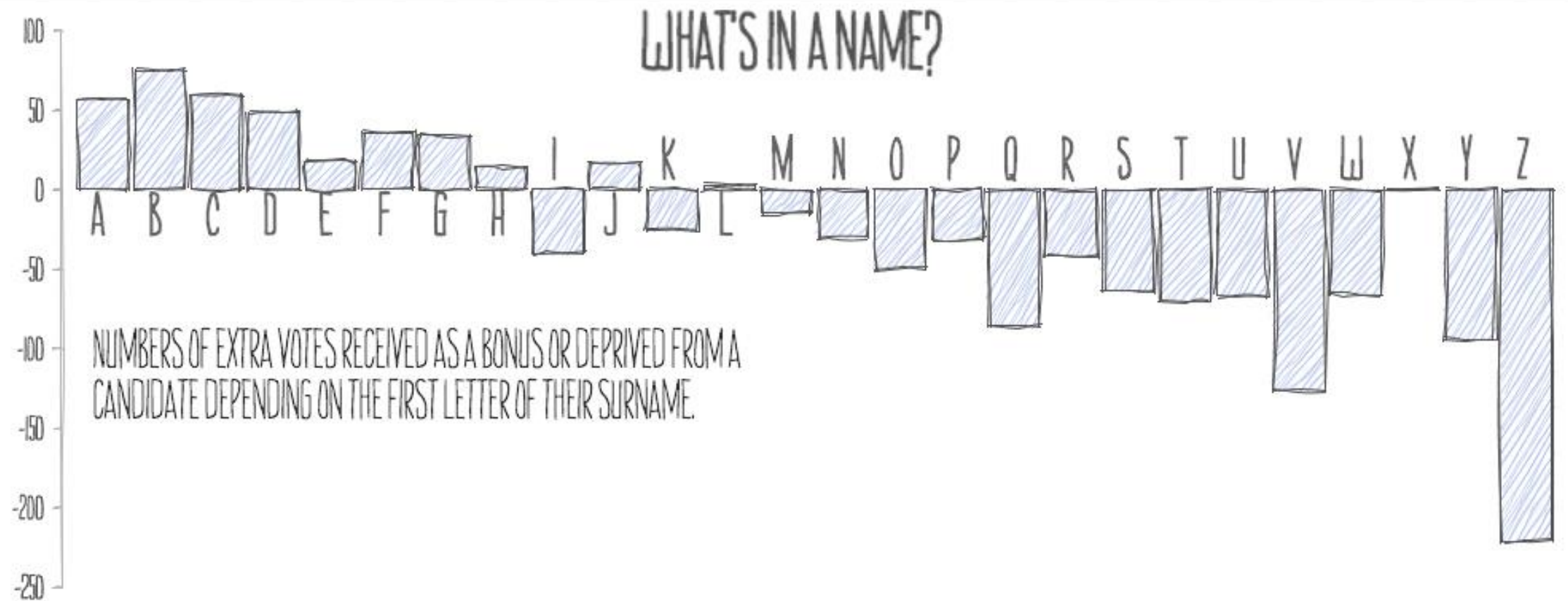
Set of discrete outcomes

- Beeswarm plot, dotplot
- Hypothetical outcome plots



Visualization Approaches

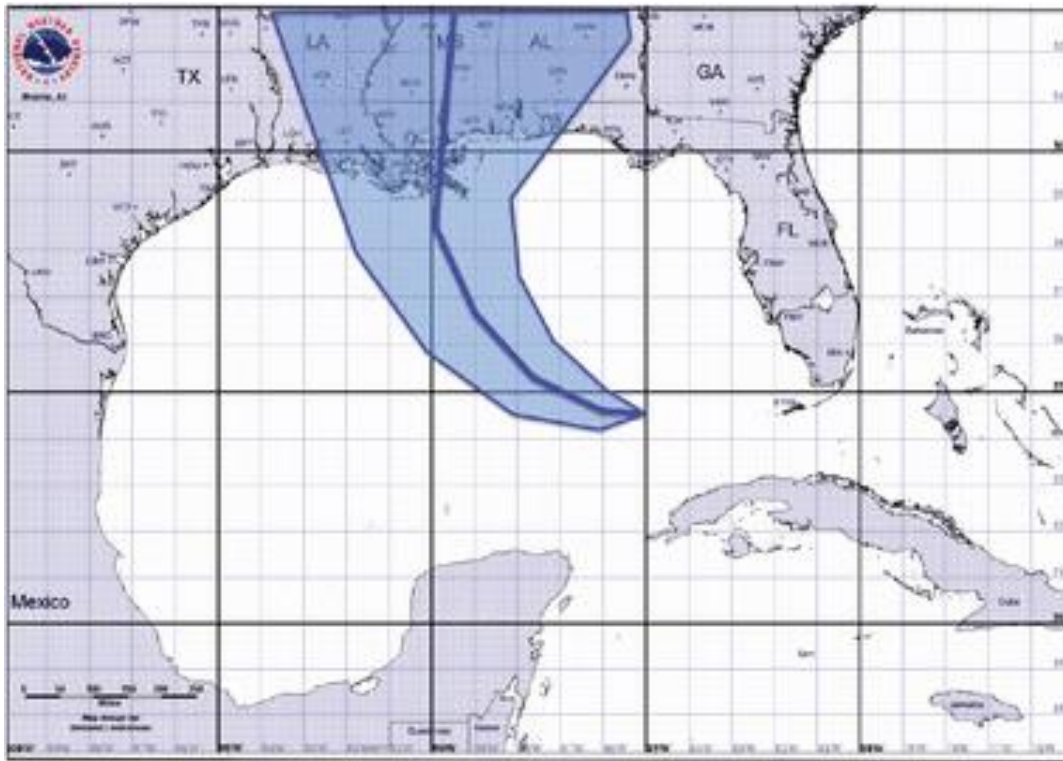
What is “sketchiness”?



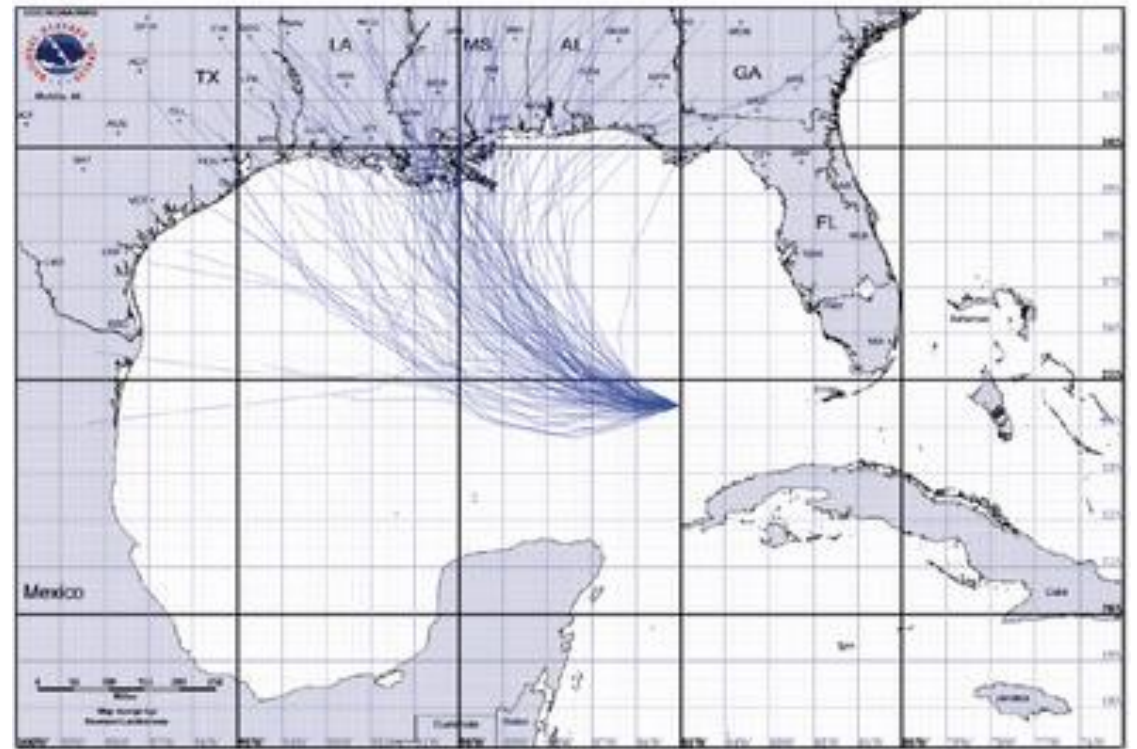
Boukhelifa et al. (2012), Wood, Jo et al. (2012)

Visualization Approaches

Summaries vs discrete outcomes



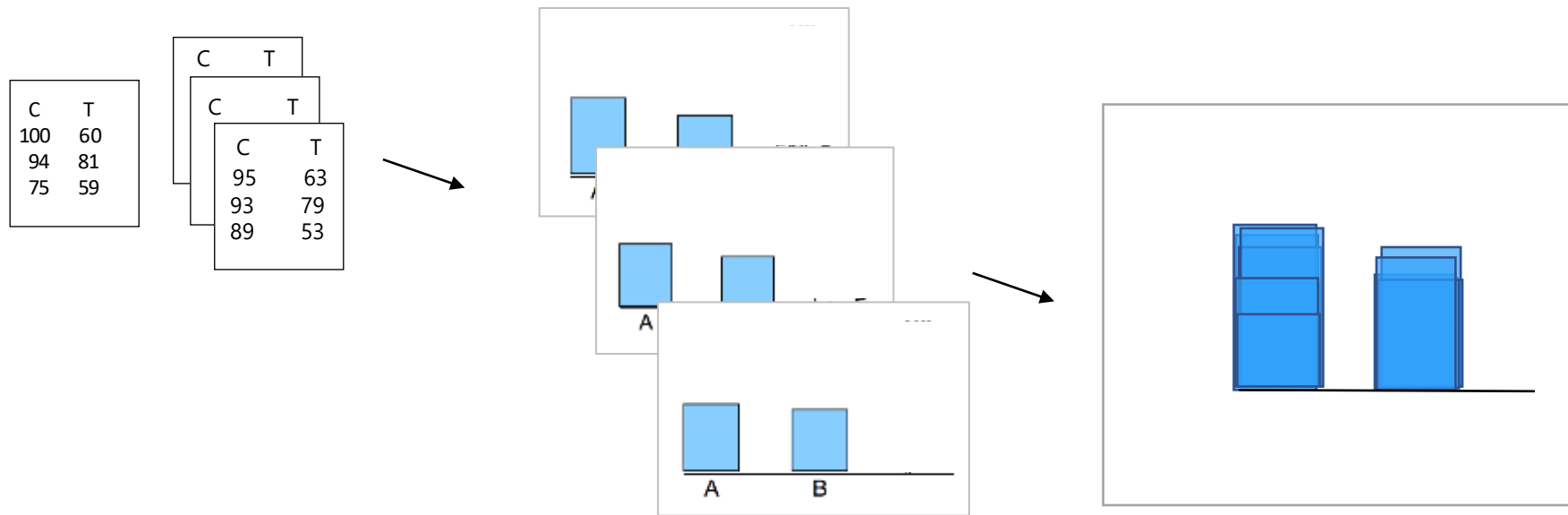
(a)



(b)

Visualization Approaches

Hypothetical Outcome Plots



Present uncertainty as samples *over time*, where each sample is a new frame in an animated visualization

Visualization Approaches

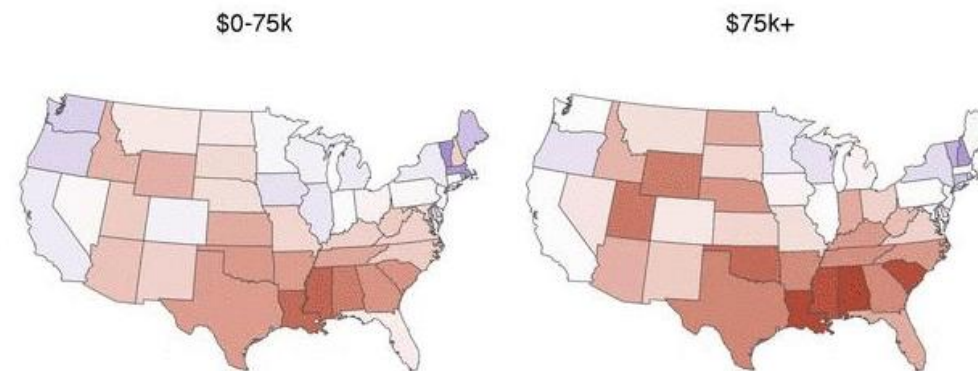
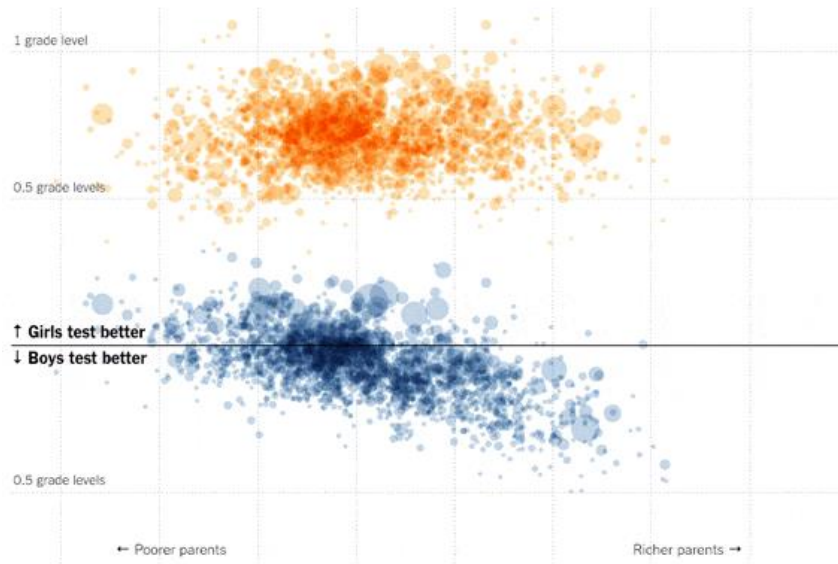
Hypothetical Outcome Plots

If those numbers are correct, sampling error could produce this debate instead...

...and leave these candidates out.

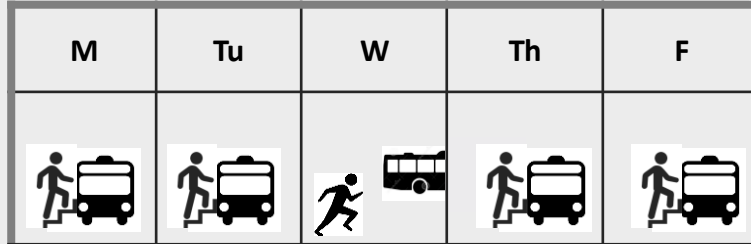


Where Boys Outperform Girls in Math: Rich, White and Suburban Districts



Applicable to arbitrary encodings

Framing Probability as Frequency



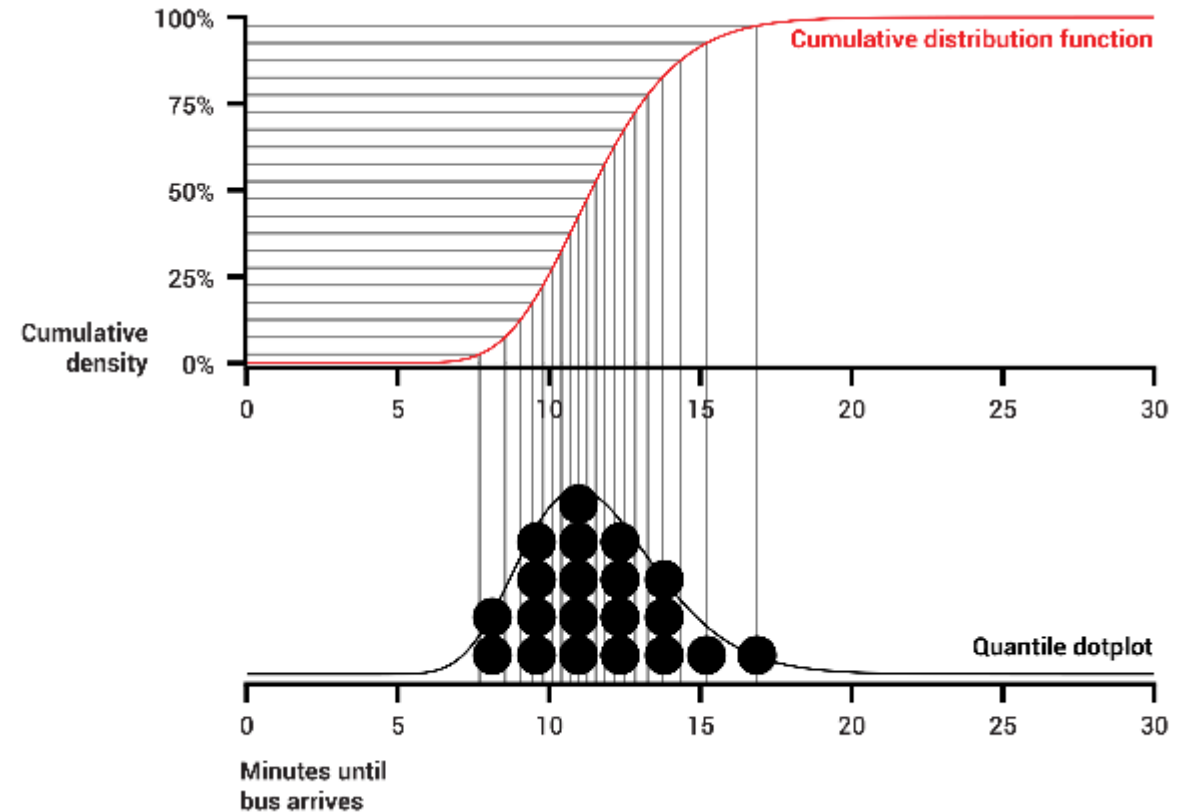
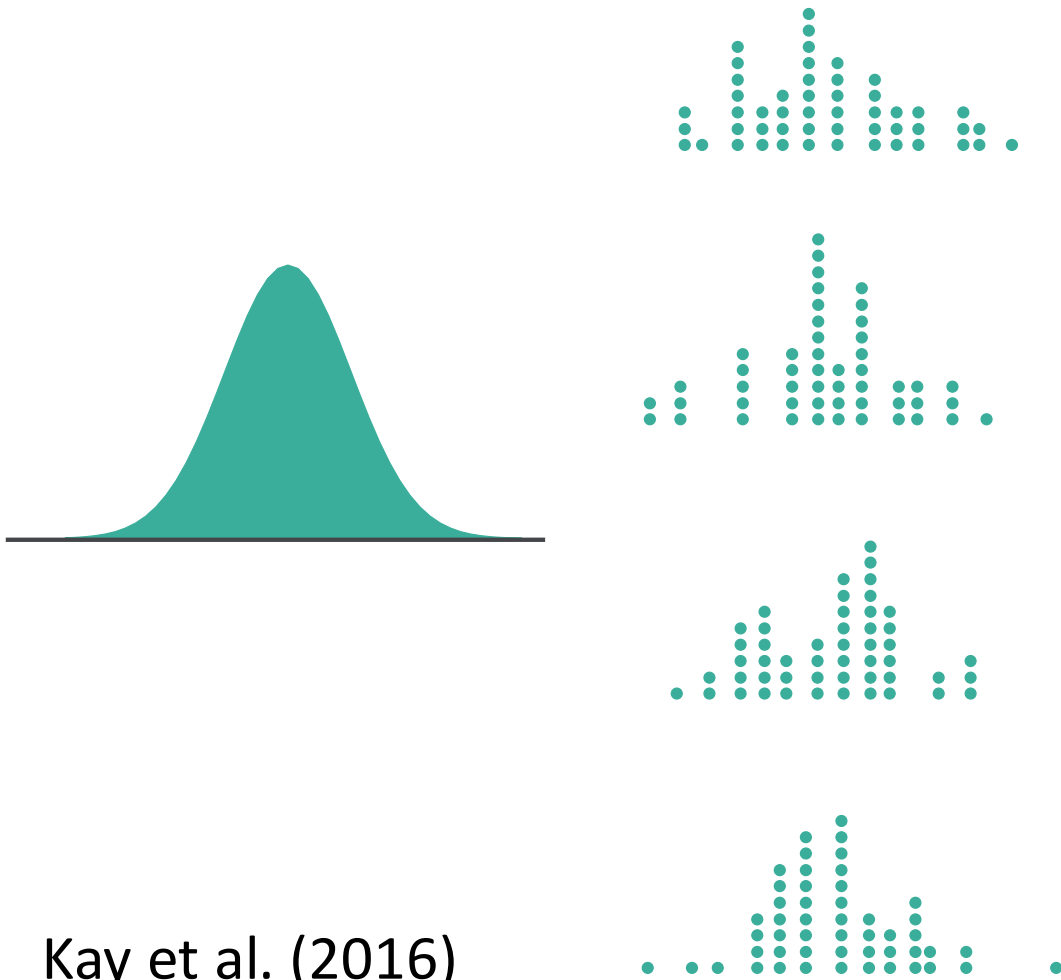
M	Tu	W	Th	F
				



Frequency framing improves reasoning
4 out of 5 times, vs 80%
(Gigerenzer & Hoffrage 1995)

Visualization Approaches

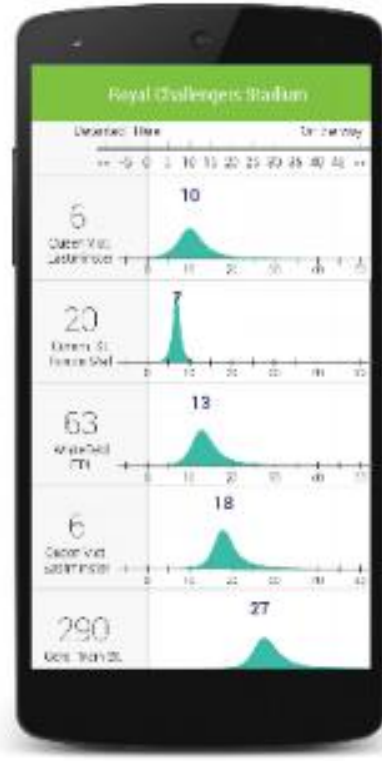
Quantile Dotplots



Kay et al. (2016)

Visualization Approaches

Quantile Dotplots



Bus riders made more accurate probability estimates with 20-dot QDPs than with 100-dot QDPs or densities

Better quality decisions with 50- and 20-dot QDPs vs densities, intervals, or text

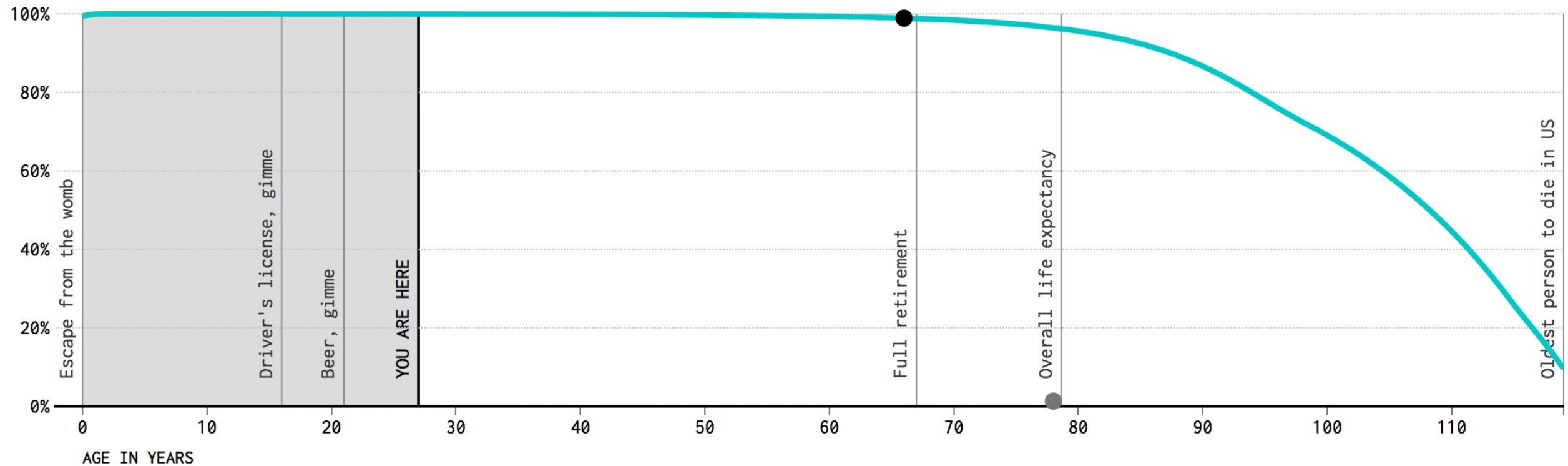
Visualization Approaches

Sedimentation

I am **female** and currently **27** years old.

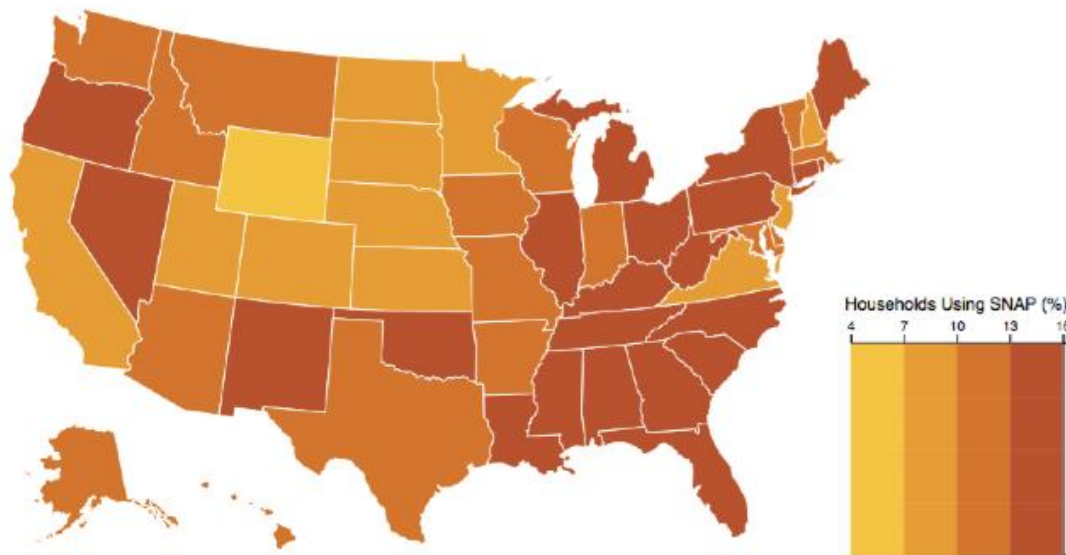
SLOW
FAST

PROBABILITY OF LIVING TO NEXT YEAR

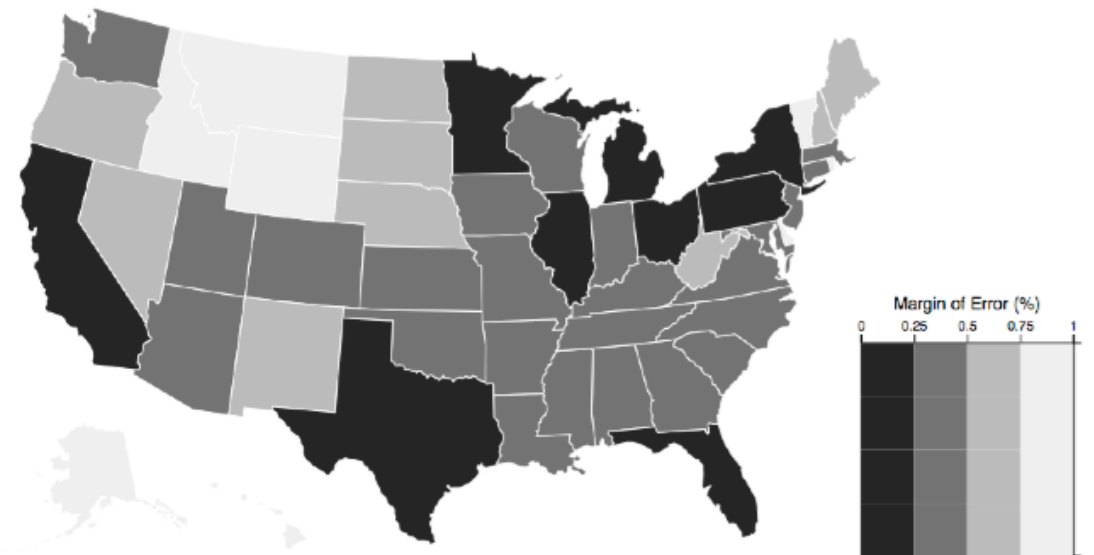


Presenting Uncertainty When Position
Encoding is Occupied

Juxtaposition

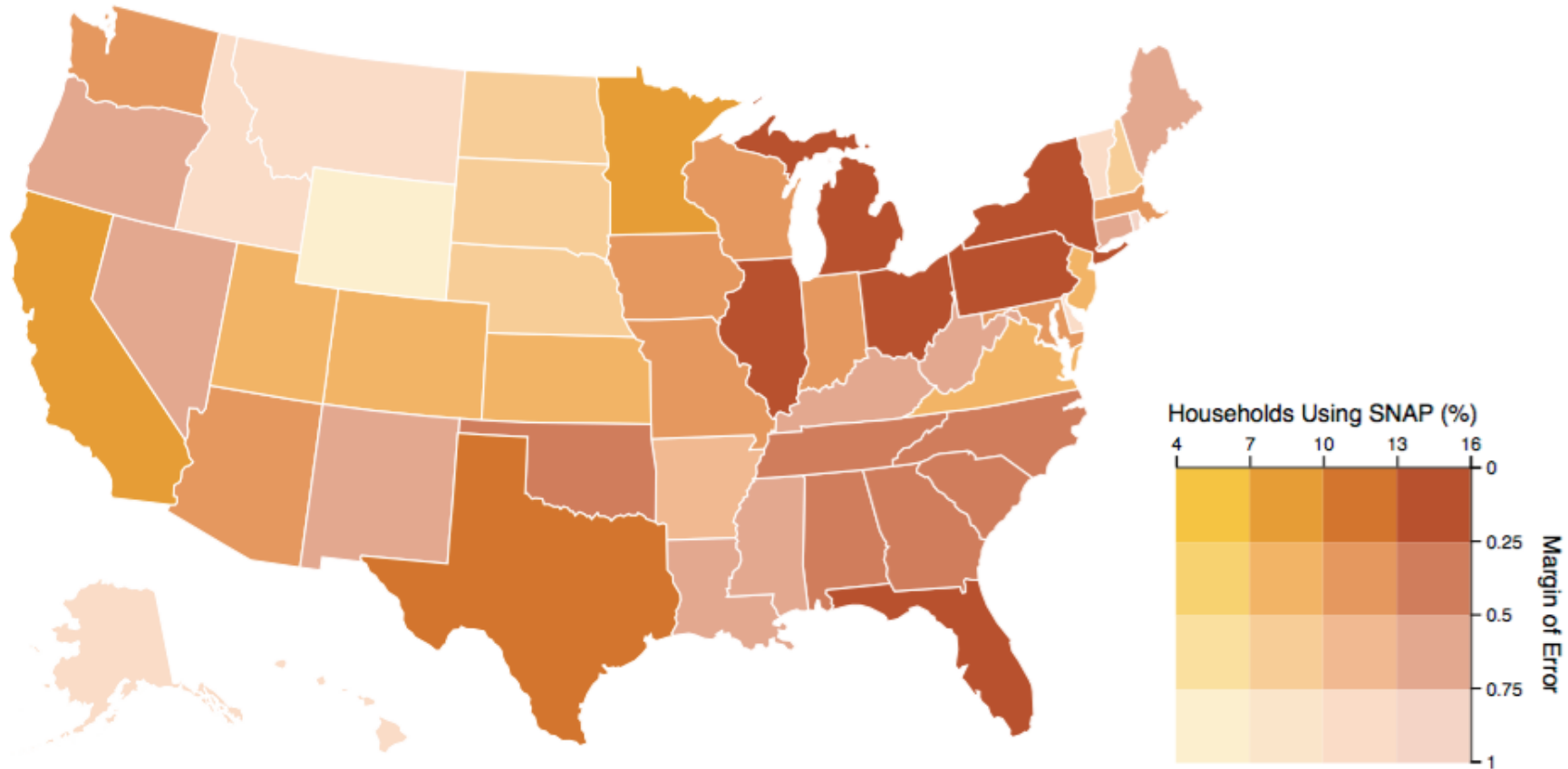


Data Map



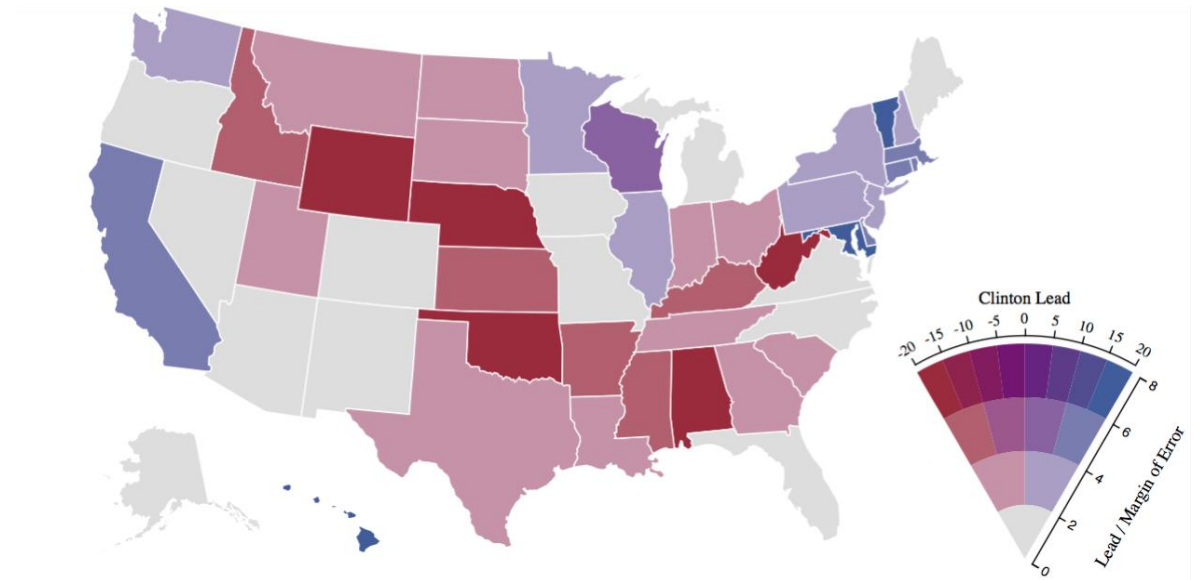
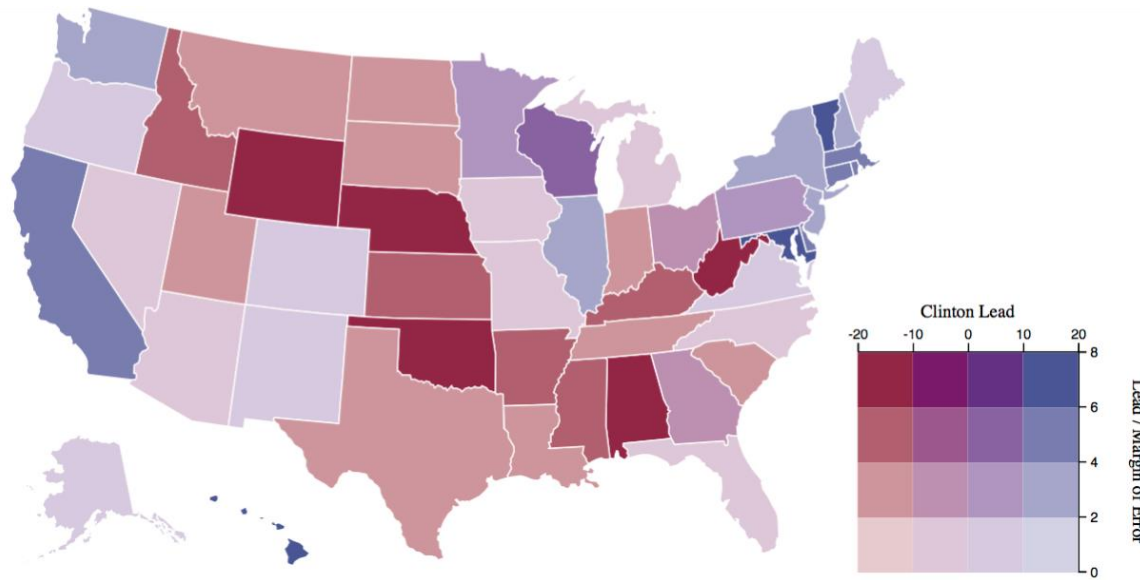
Uncertainty Map

Superposition



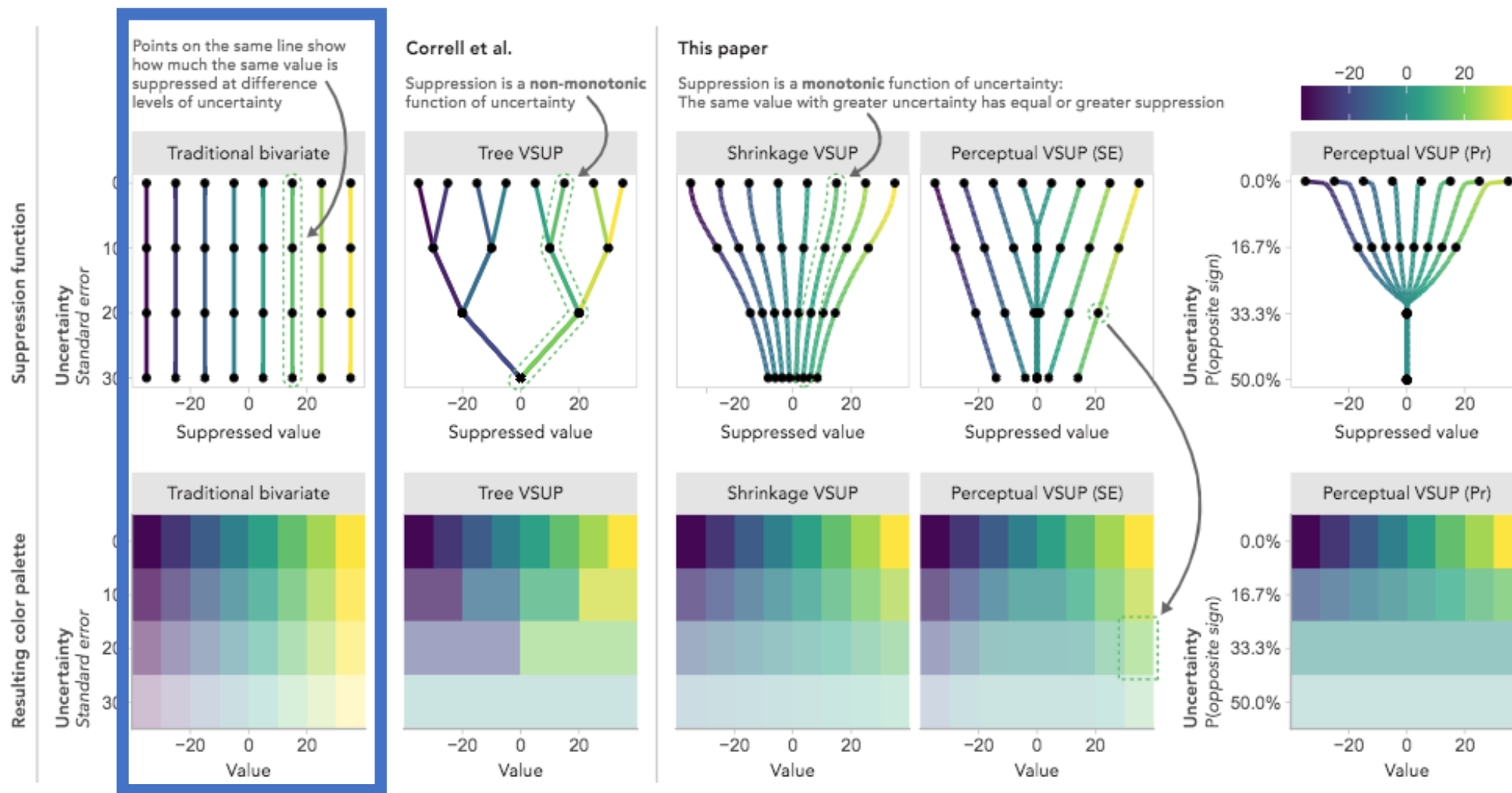
Bivariate Map

Value-Suppressing Uncertainty Palettes



Correl et al. (2018)

Value-Suppressing Uncertainty Palettes



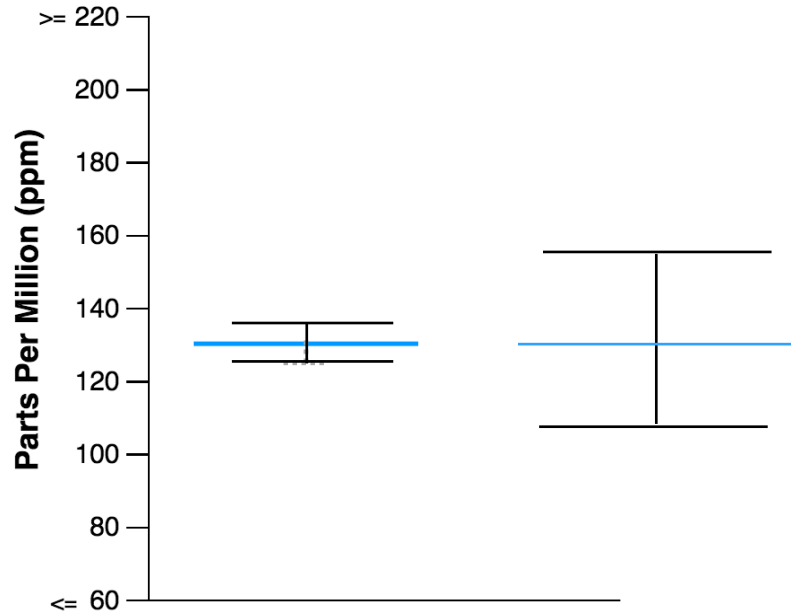
Kay (in press)

Evaluating Uncertainty Visualizations

“The advantage for uncertainty estimates depends critically on how they are expressed. It is crucial that the expression is compatible with both the decision task and cognitive processes of the user.”

Joslyn & LeClerc (2013)

Does the vis support the user's task?



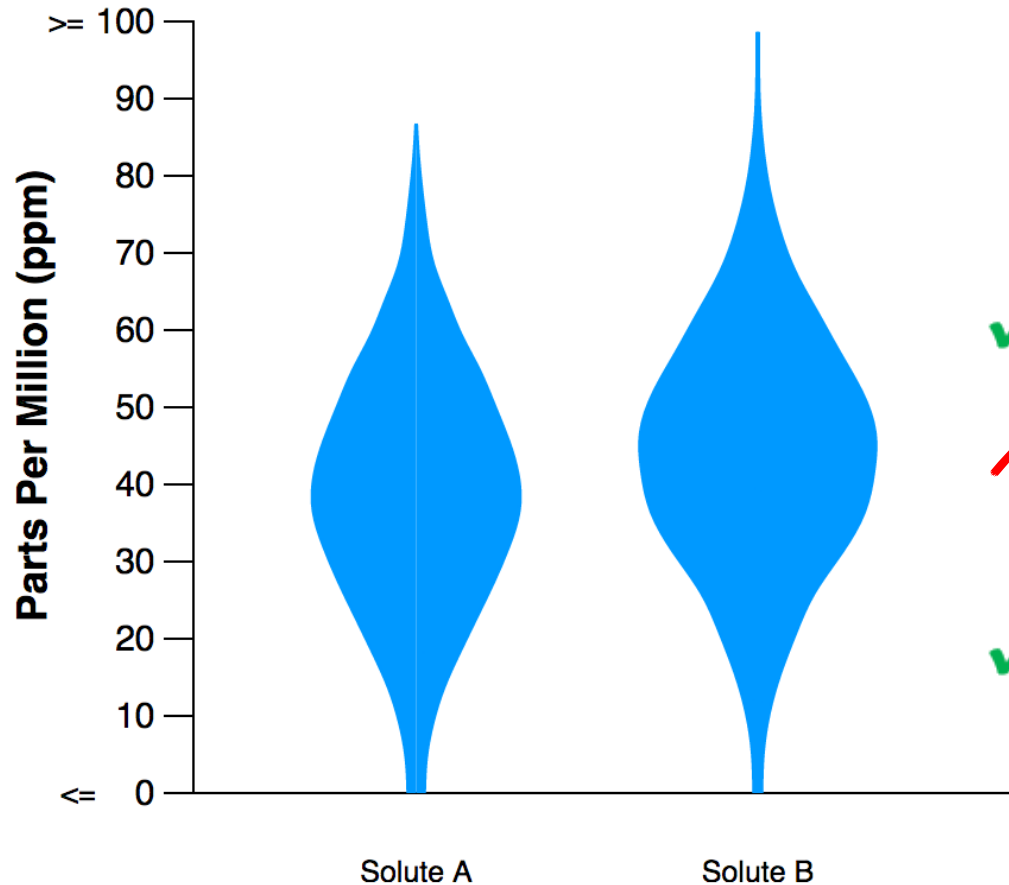
Is the model of uncertainty appropriate to answer the user's question?

What is the level in the entire ocean?

vs

How much do the readings vary?

Does the vis support the user's task?



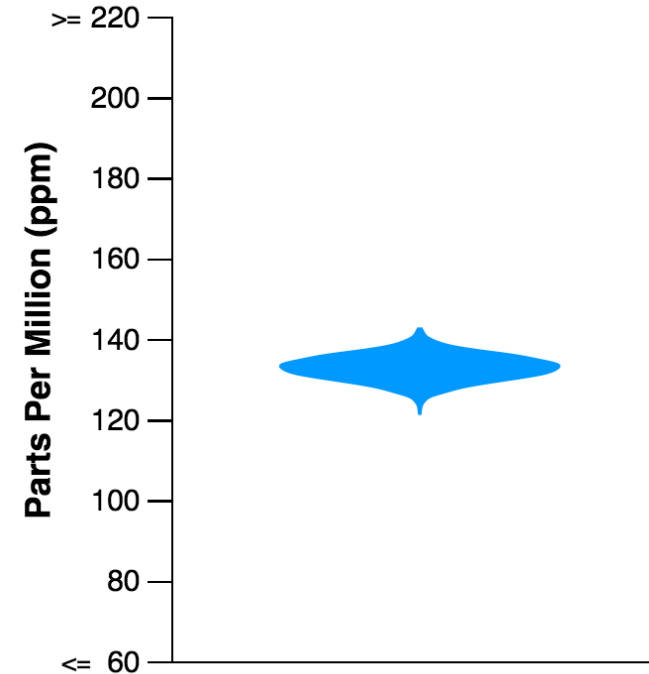
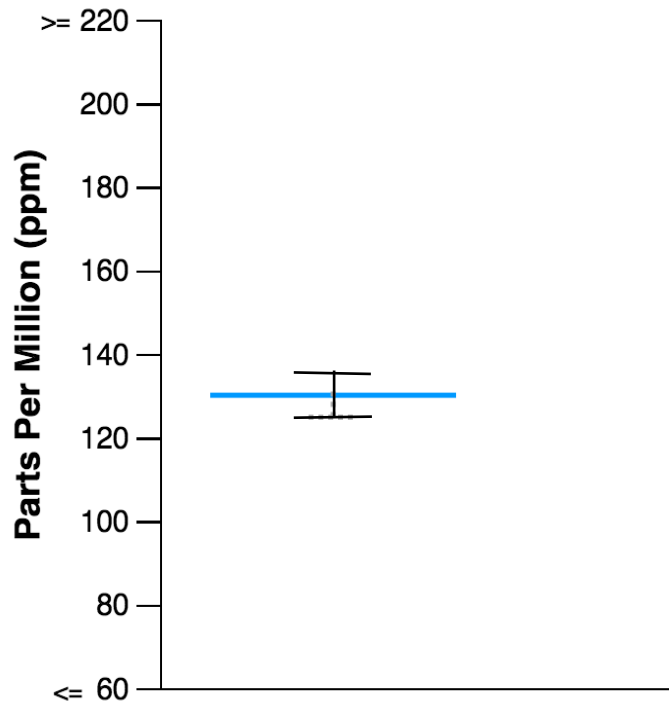
Can the user extract the needed information from the visualization?

- ✓ *How confident are you that $\mu_B > \mu_A$?*
- ✗ *What is probability that $B > A$?*
- Are A and B correlated?*
- ✓ *Which distribution has larger variance?*

Is the vis expressive & can users infer meaning?

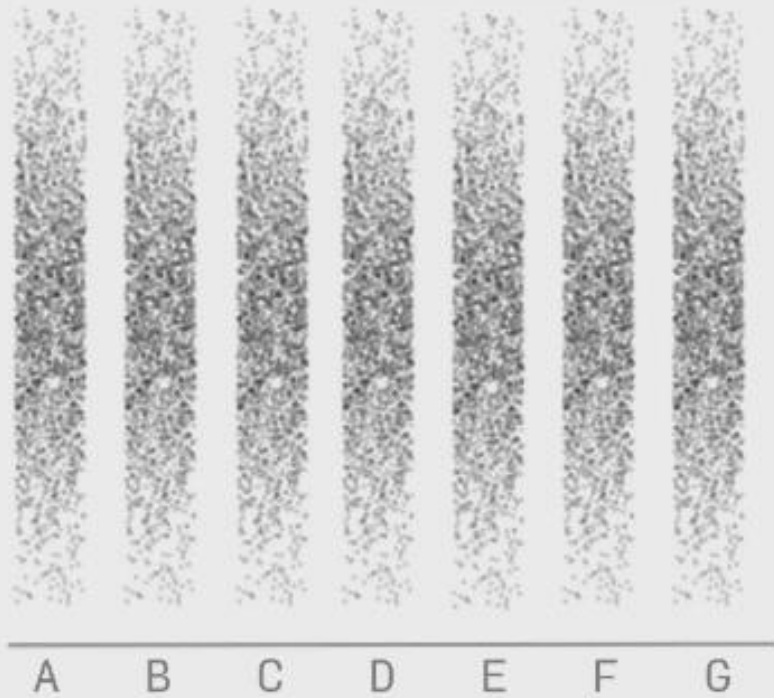
Expressiveness: How well does the visualization present all the facts (and only the facts)?

Mackinlay (1986)

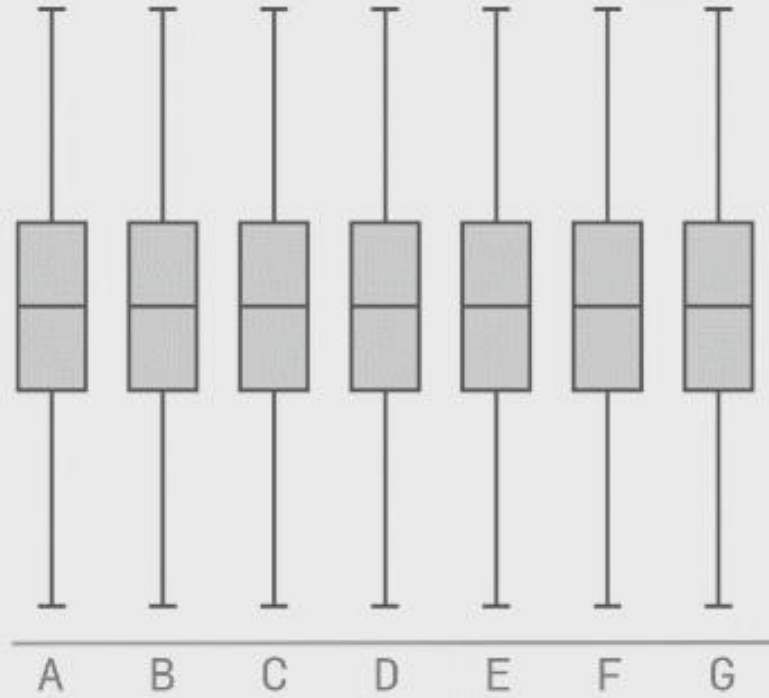


Is the vis expressive & can users infer meaning?

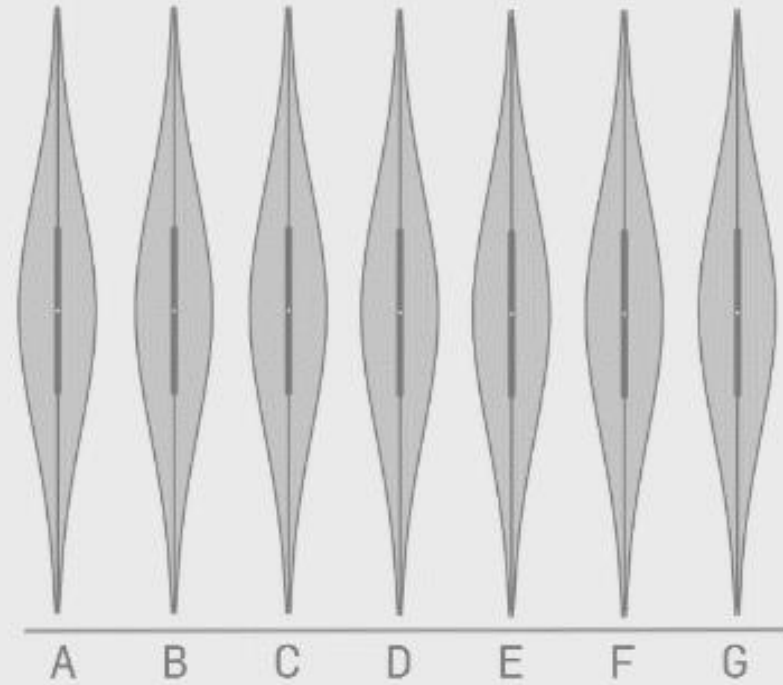
Raw Data



Box-plot of the Data

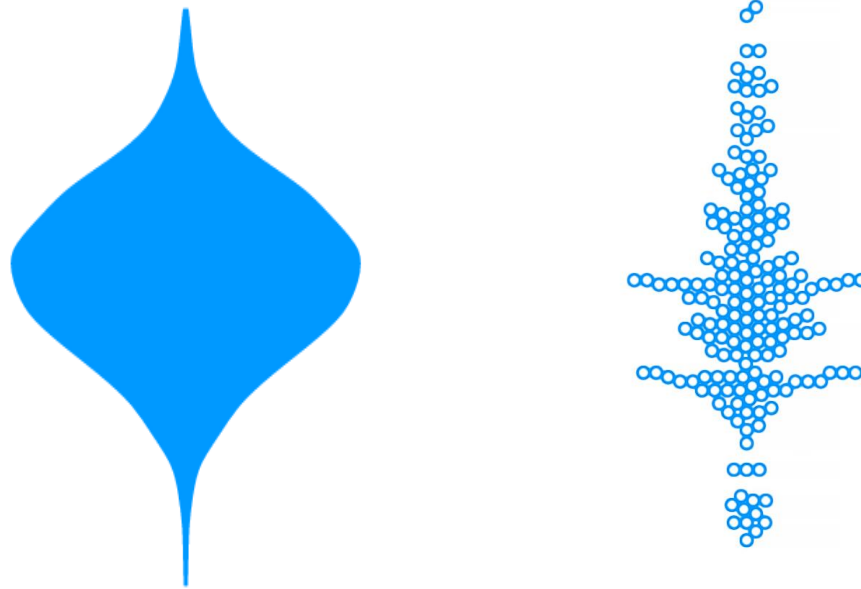


Violin-plot of the Data



What is the information format?

Does the visualization present data in a way that aligns with human reasoning abilities?



Can users decode the data from the chart?

Effectiveness: How easily can a human estimate the values from the visualization?

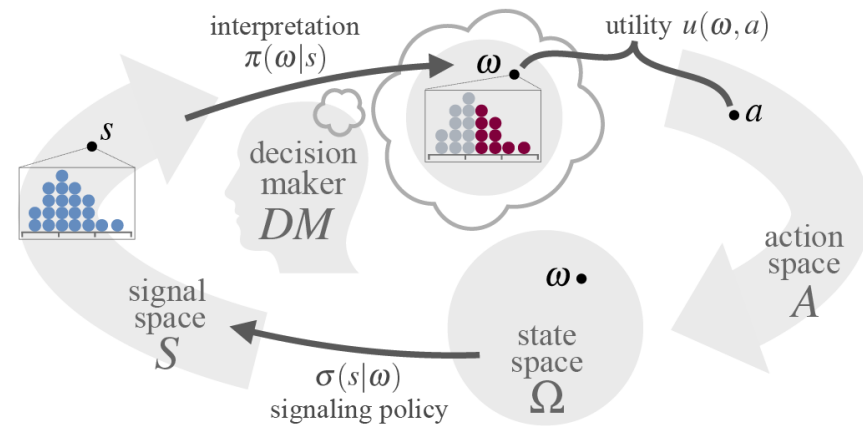
Mackinlay (1986)



Quantitative		Ordinal		Nominal
Position	_____	Position	_____	Position
Length		Density		Hue
Angle		Saturation		Texture
Slope		Hue		Connection
Area		Texture		Containment
Volume		Connection		Density
Density		Containment		Saturation
Saturation		Length		Shape
Hue		Angle		Length
Texture		Slope		Angle
Connection		Area		Slope
Containment		Volume		Area
Shape	_____	Shape		Volume

Goal: Robust visualization for inference

Decision making



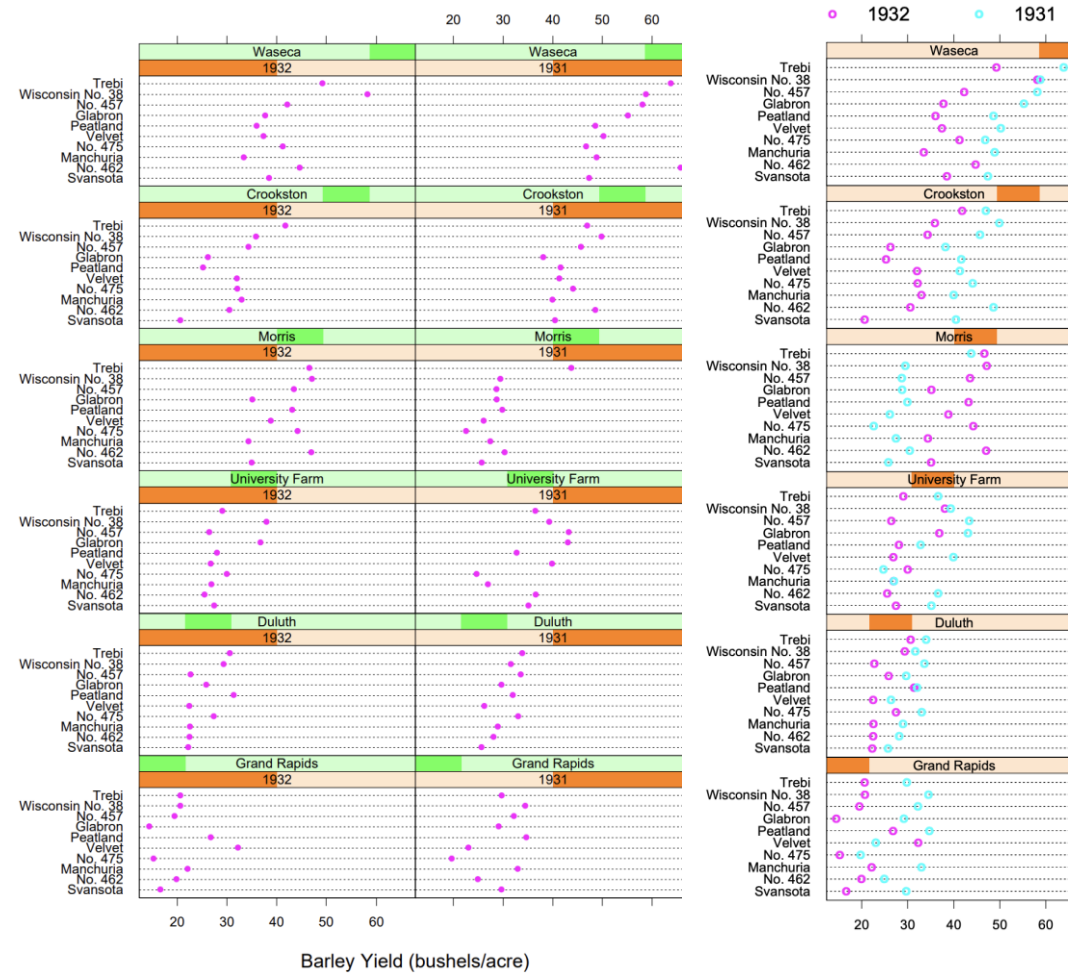
Model checking



Predictability

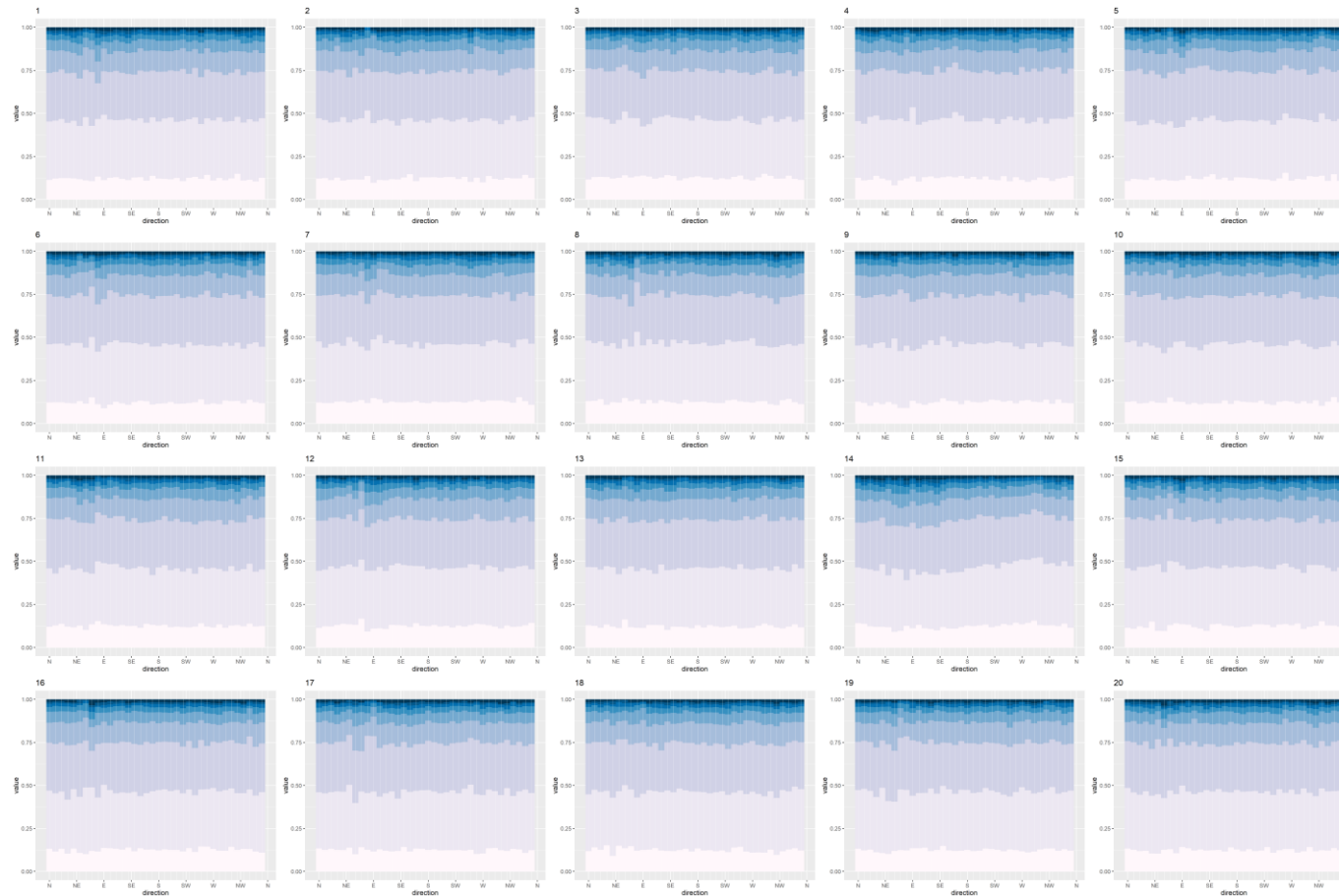


Relationship between models and charts



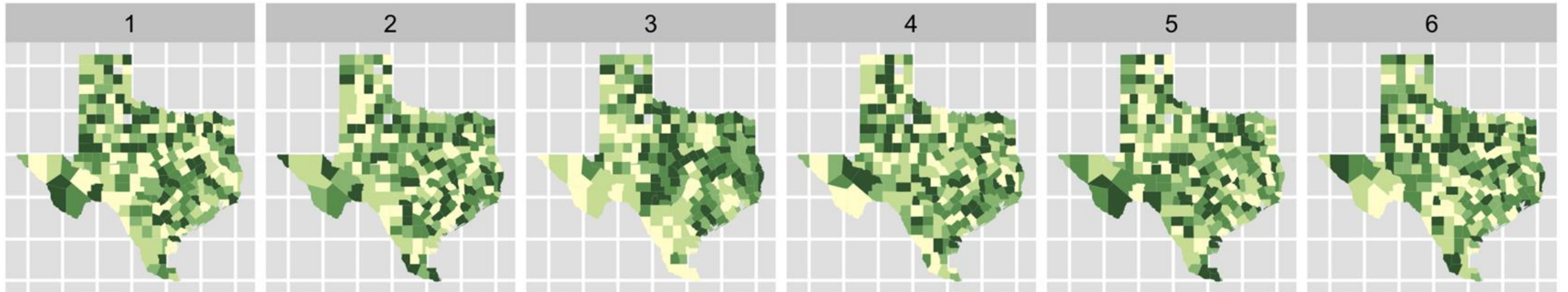
The visualization “line up”

Can you spot the real data?



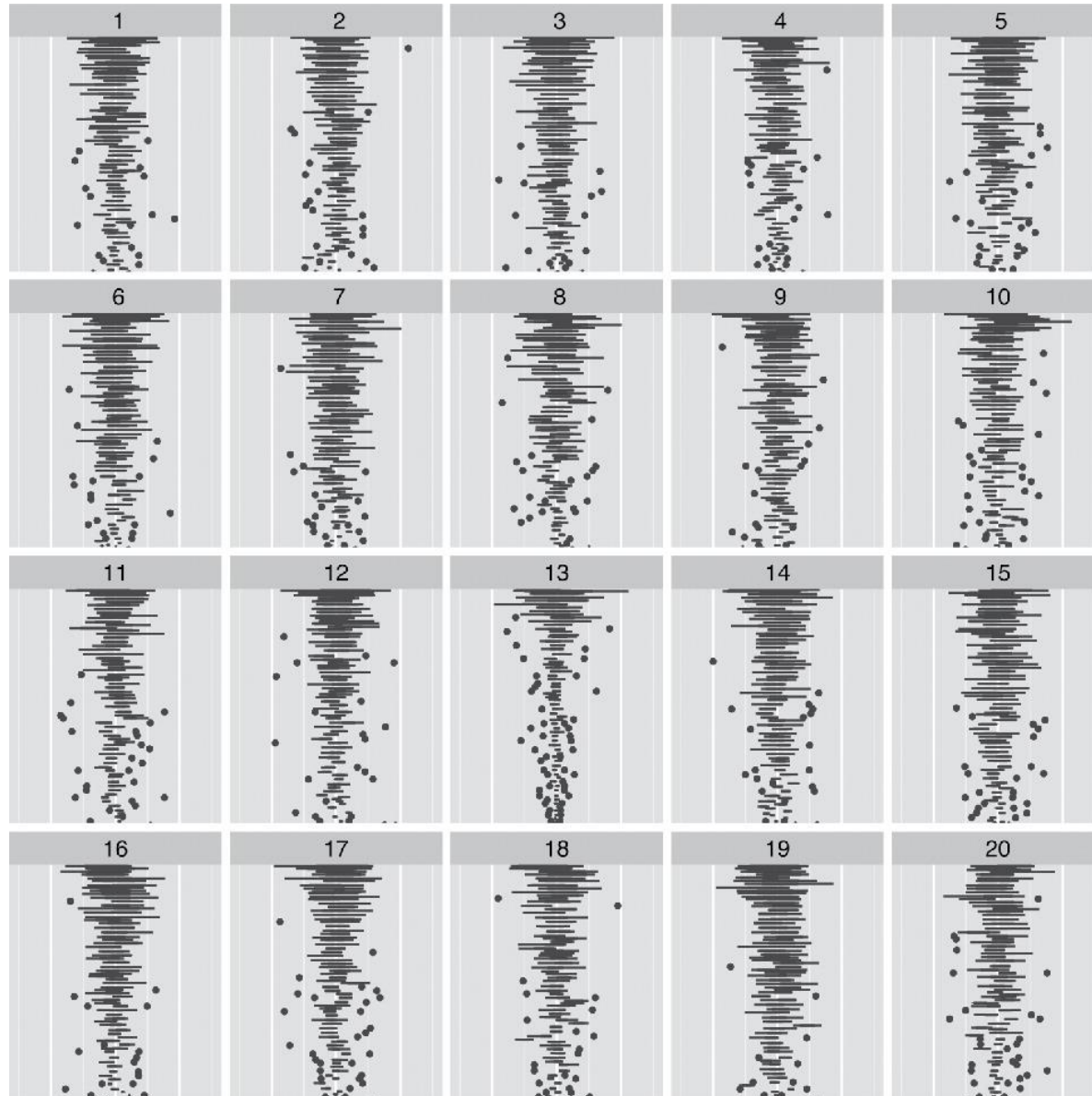
The visualization “line up”

How we define a “null model” is super important!

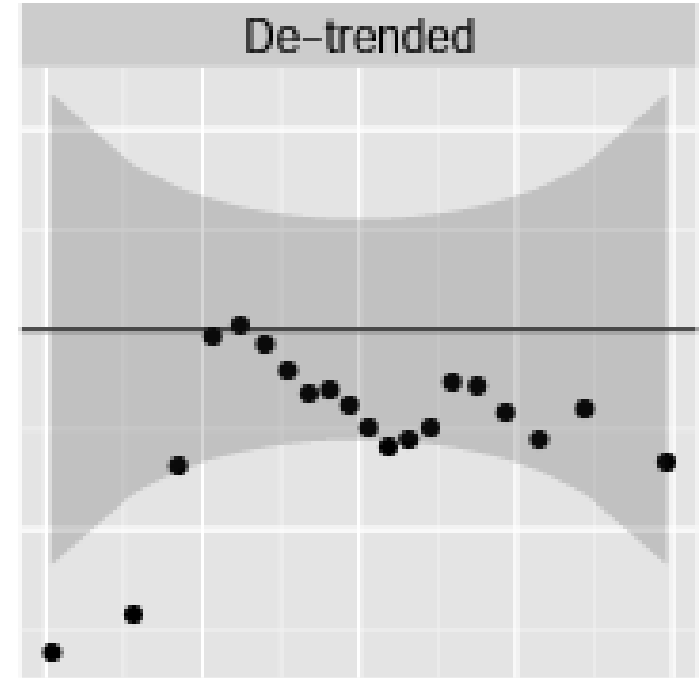
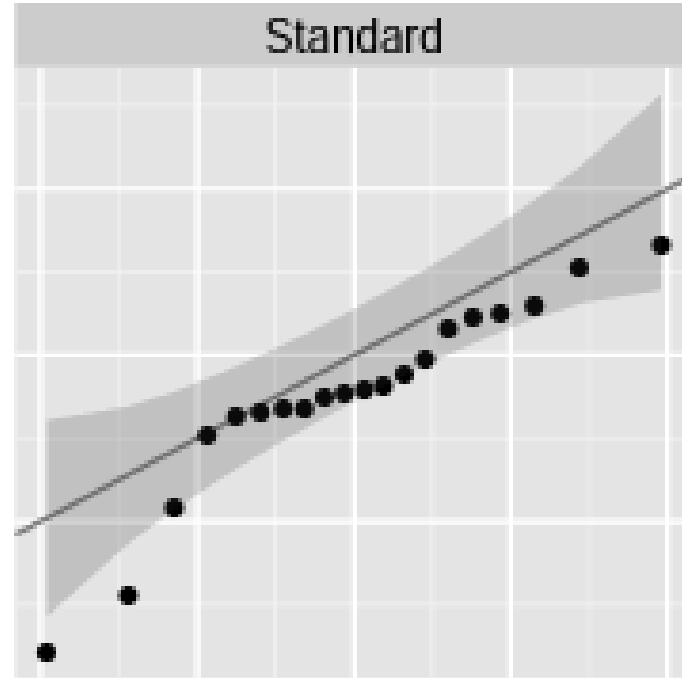
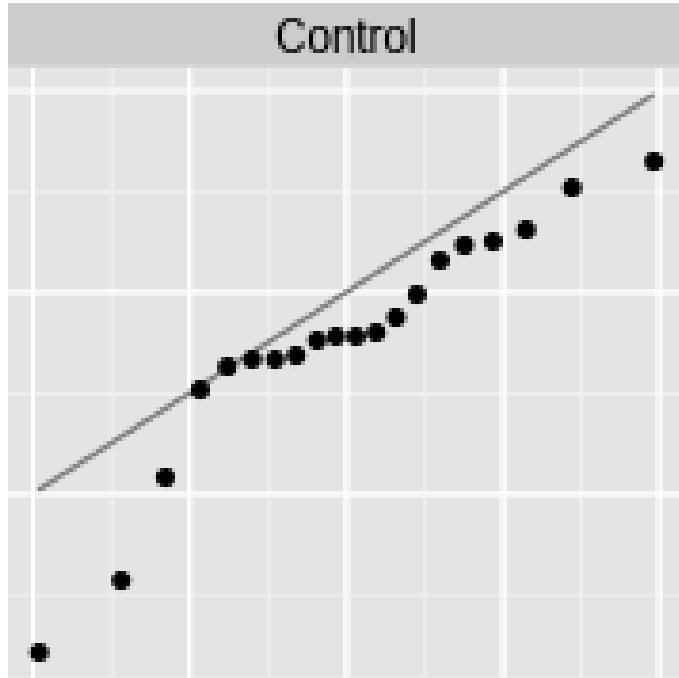


Wickham et al. (2010)

The visualization “line up”

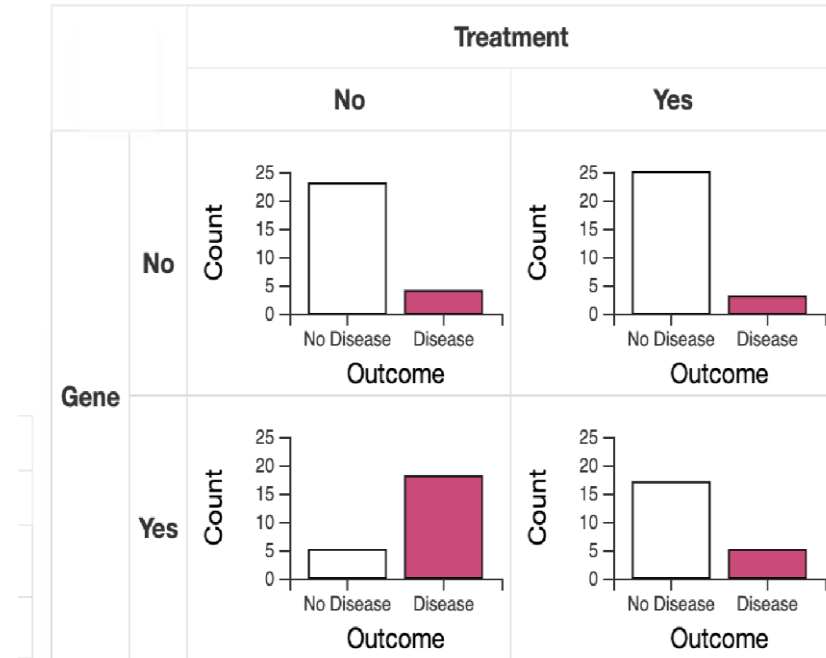
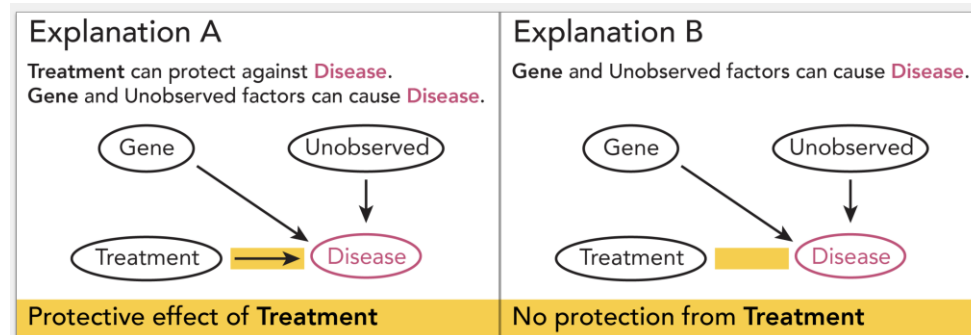


QQ-Plots



Loy et al. (2014)

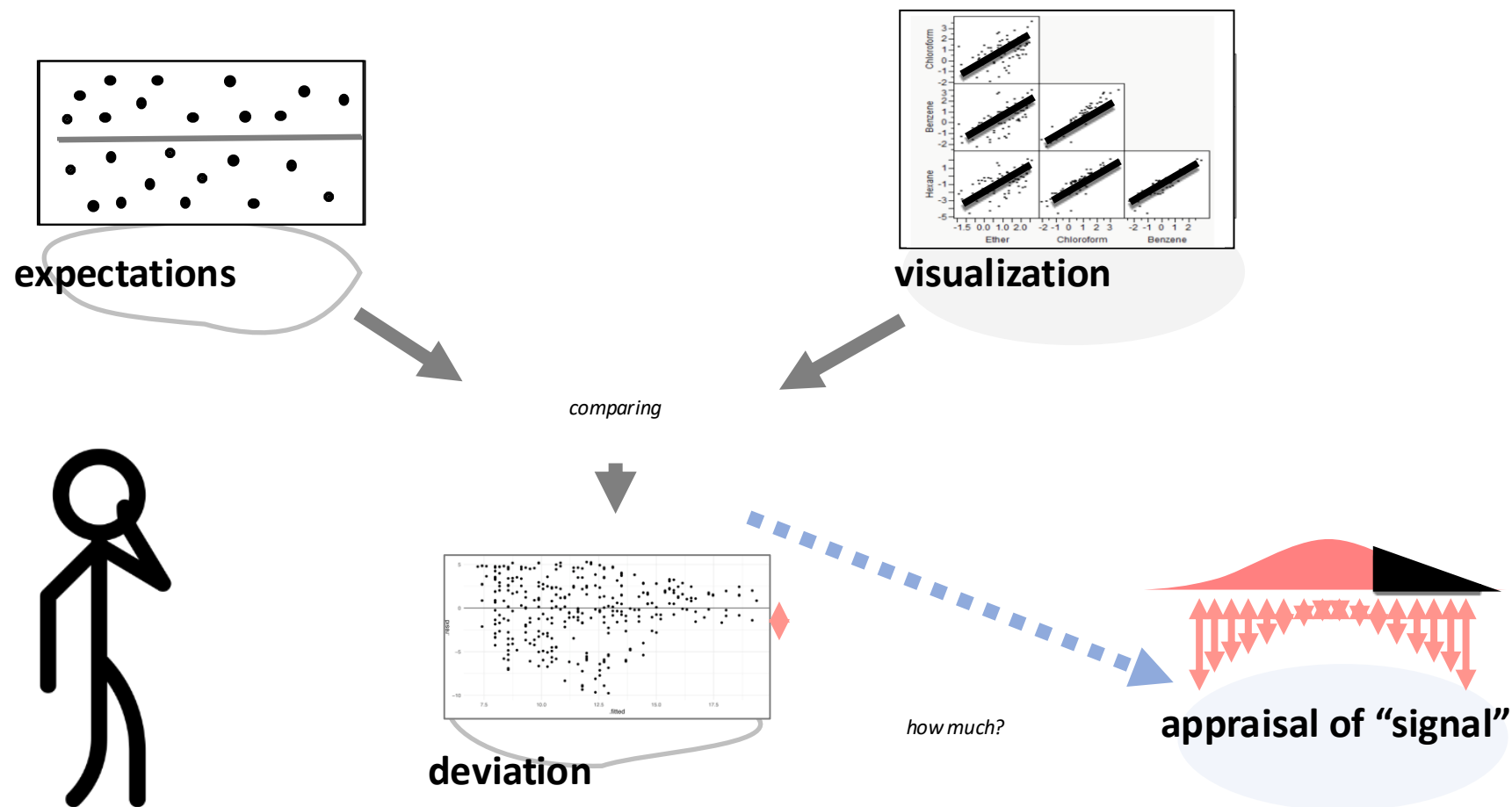
Imagining counterfactuals can be hard



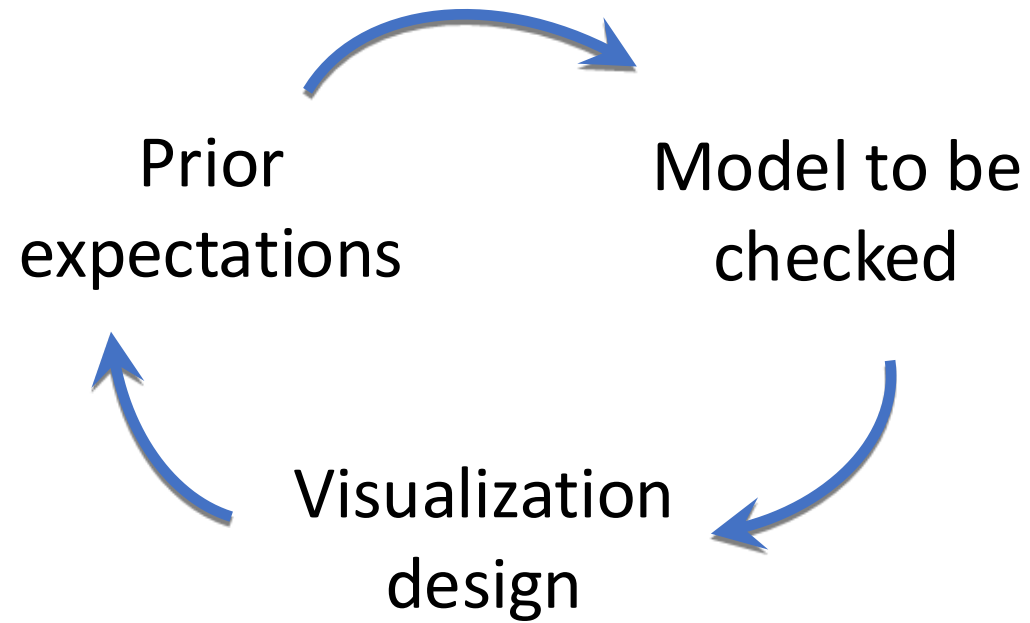
How much probability (out of 100) would you assign to each explanation?

- Insensitivity to sample size.
- More sensitive to falsifying information.
- Visualization didn't help much.

Implicit model checks give charts meaning



Idealized role of visualization in inference



Thanks!

Alex Kale

<https://people.cs.uchicago.edu/~kalea/>



@AlexKale17



kalealex

Slides adapted from Jessica Hullman, Michael Correl, & Arvind Satyanarayan

Presenting Uncertainty for Multivariate Data

Multivariate Data

Example: MR spectroscopy

Measure the resonance frequencies of molecules in the brain.

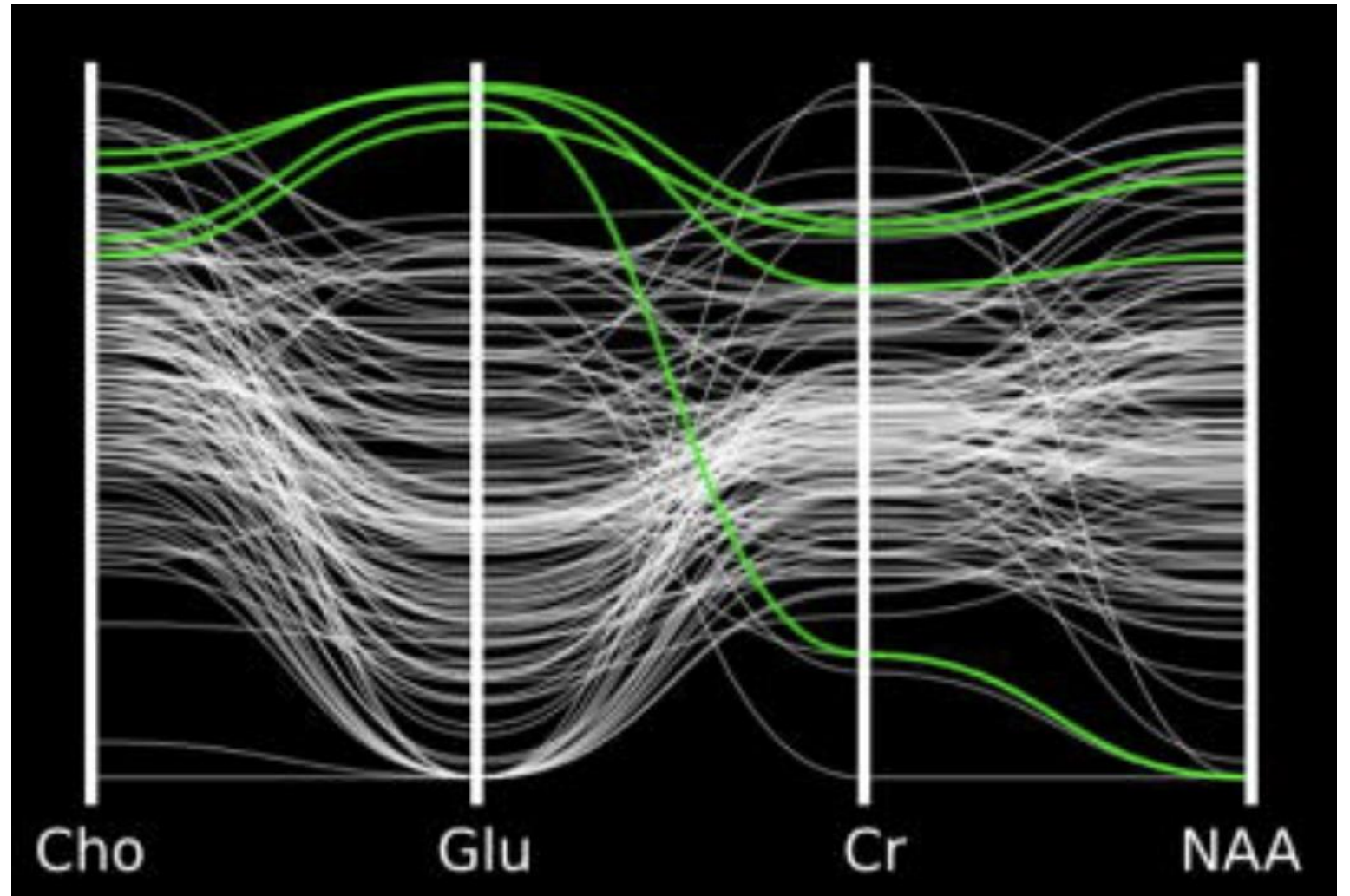
Voxels: spatial resolution of about a cubic centimeter

Model concentrations of different molecules.

Data format: One row per voxel; One column per molecule.

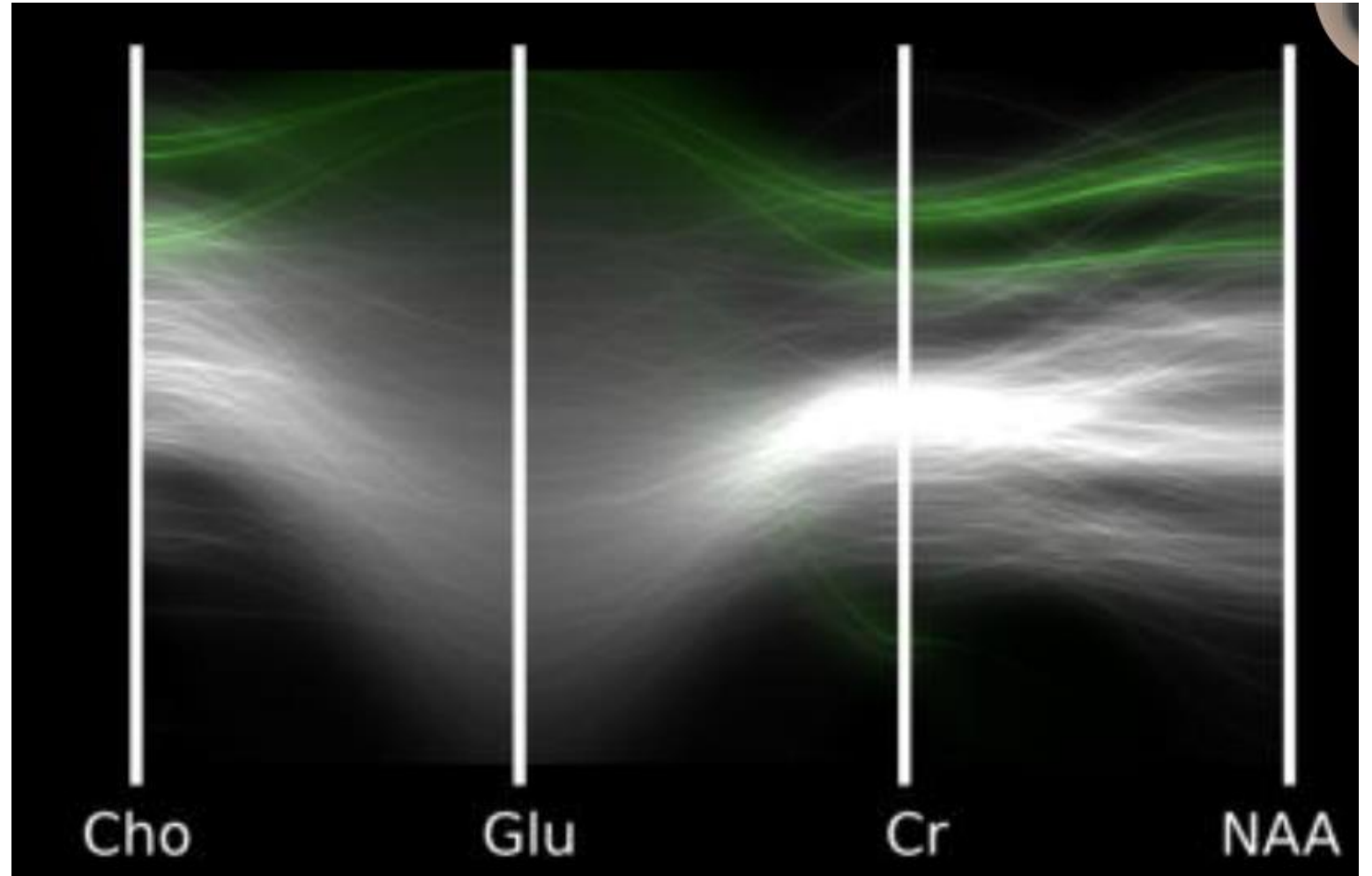
Parallel Coordinates

One line per voxel
(spatial unit)



Parallel Coordinates for Distributions

Show multivariate
distribution for each
voxel



Step 1: Interpolation

Input: Two distributions

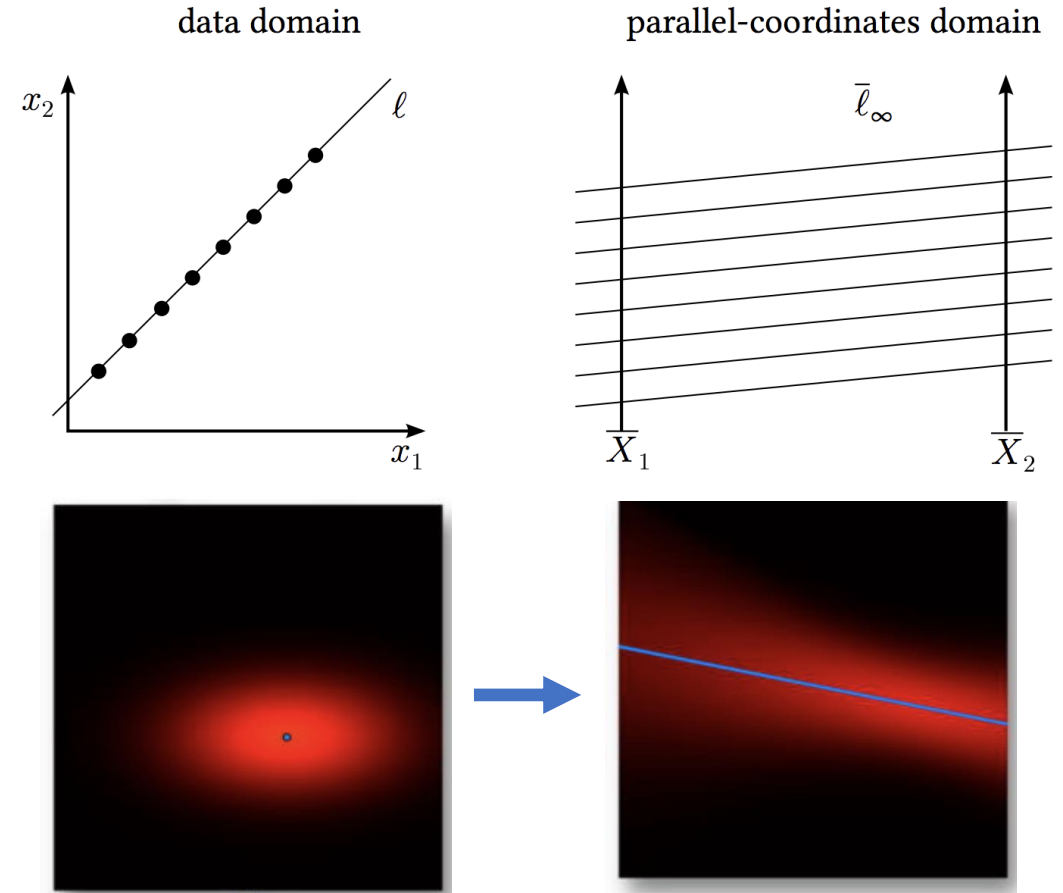
$$x_1 = N(\mu_1, \sigma_1^2)$$

$$x_2 = N(\mu_2, \sigma_2^2)$$

Output: μ and σ^2 across
horizontal axis $a \in [0, 1]$

$$\mu_a = (1 - a)\mu_1 + a\mu_2$$

$$\sigma_a^2 = (1 - a)^2 \sigma_1^2 + a^2 \sigma_2^2$$

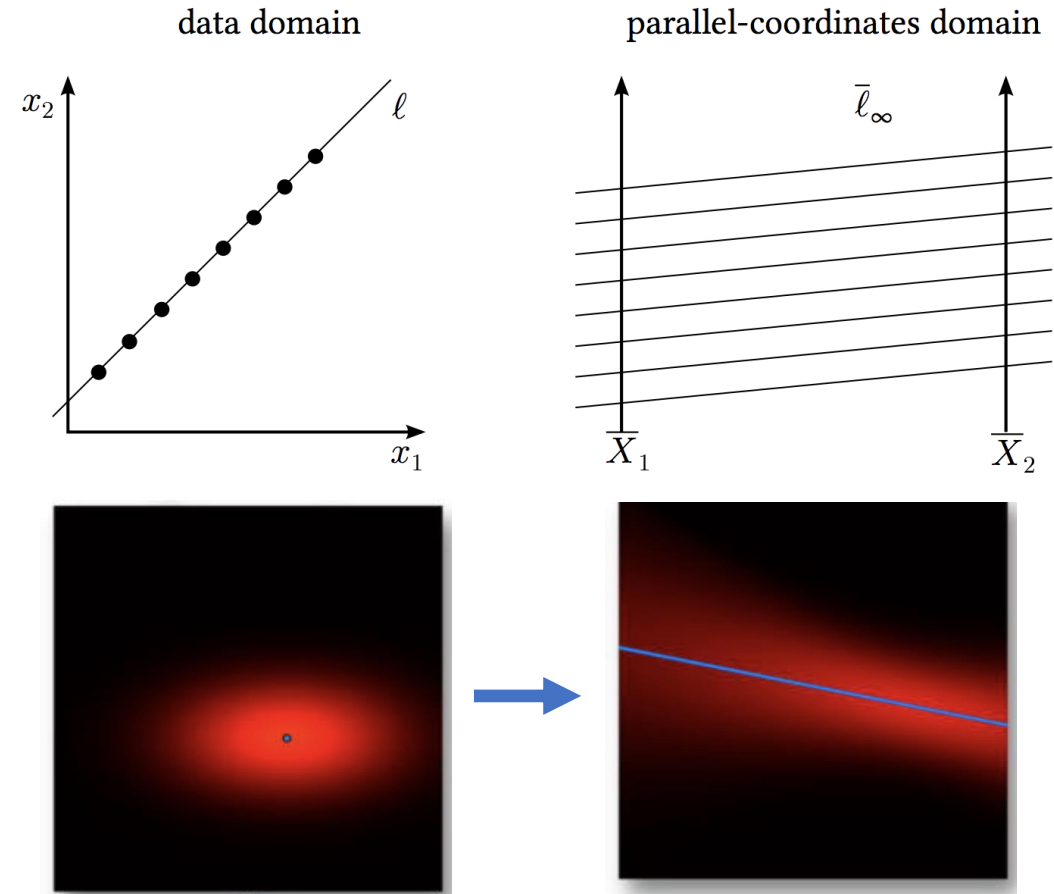


Step 2: Calculate Probability Density

Input: μ_a and σ_a^2

Output: Probability density
over horizontal span a and
vertical span b

$$PC(a, b) = (1/\sqrt{2n\sigma_a^2})\exp[-(b-\mu_a)^2/2\sigma_a^2]$$



Emphasizing Means

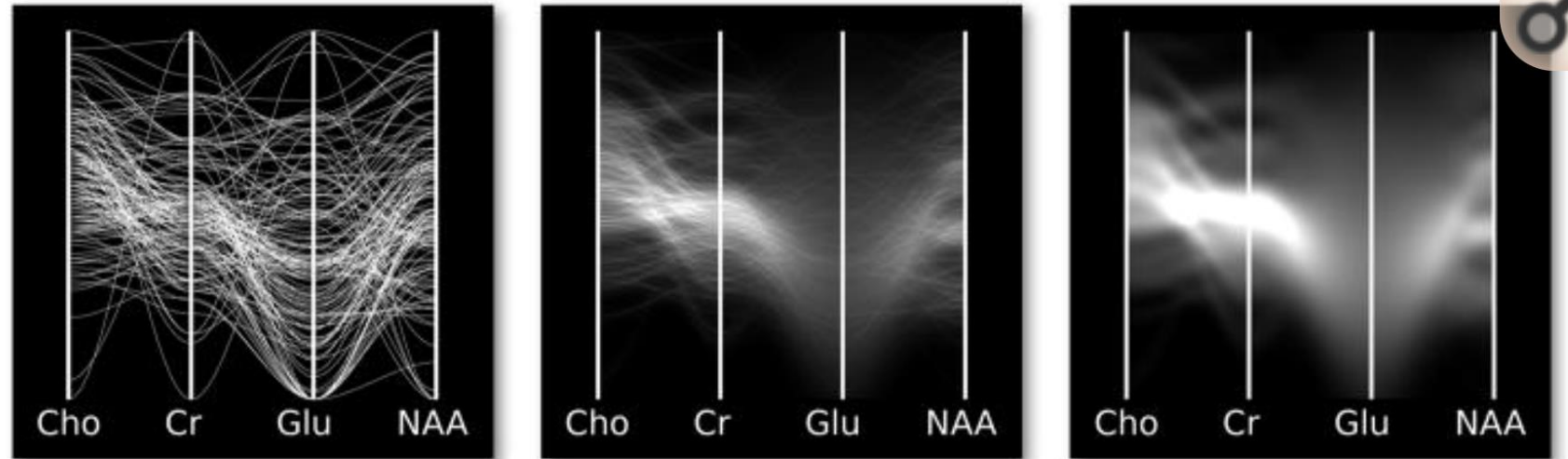
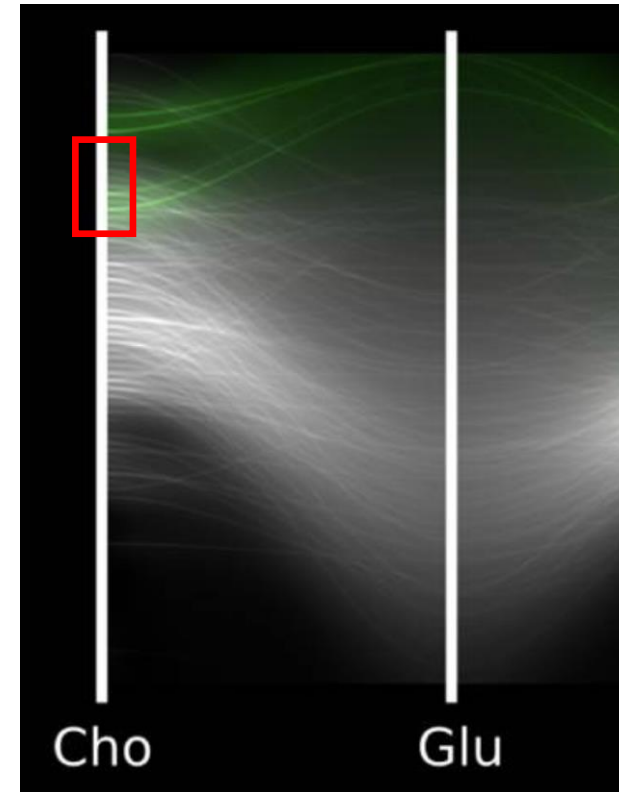
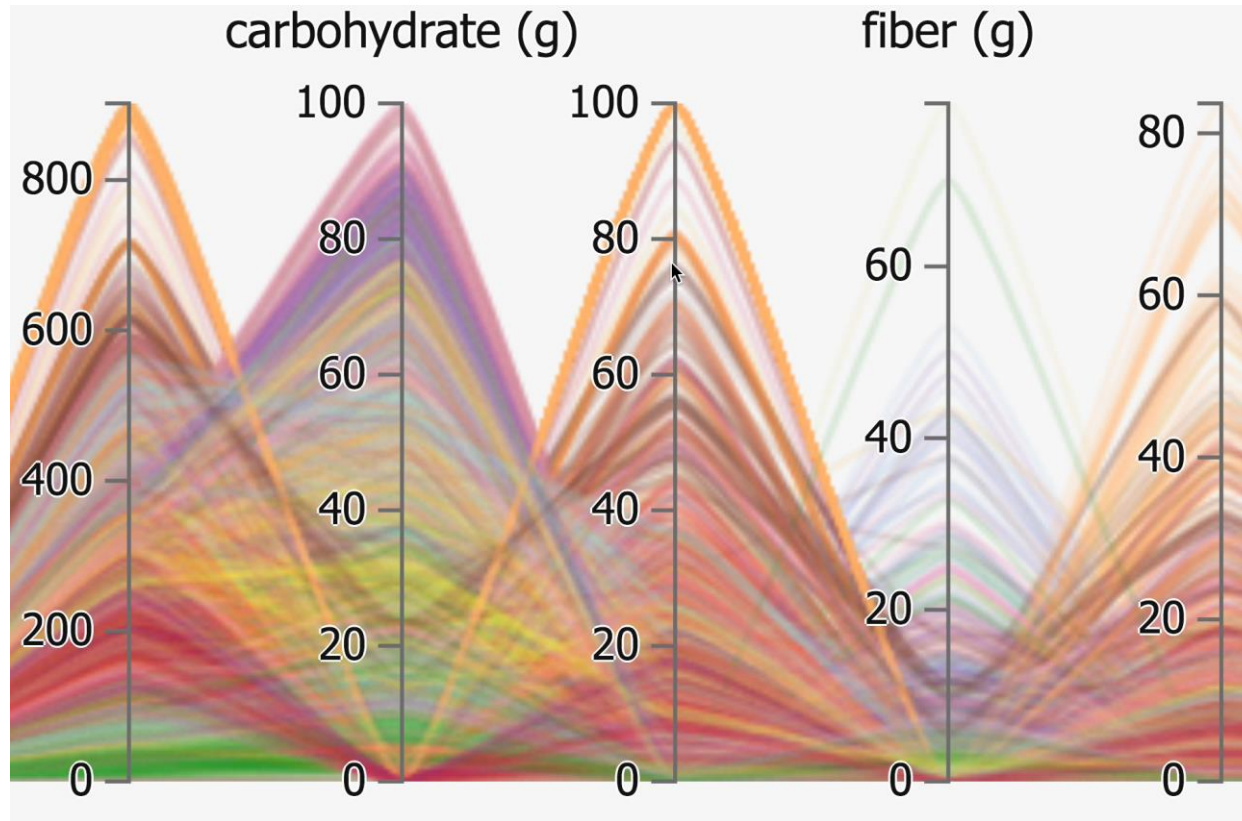


Fig. 6

Parallel coordinates plots of four MR spectroscopy metabolites. From left to right, means are decreasingly emphasized. Left: a sigmoidal PC plot of the same data shown in [Figure 4](#) with two additional variables (glutamate and n-acetylaspartate). Center: the estimated PDF mapped into PC space, with means emphasized according to their uncertainty. Right: direct visualization of the PDF.

Implications for Brush Selection



Compute integral within range of selection

Define threshold (e.g., 95%) to determine what lines are selected

Limited for Correlated Variables

